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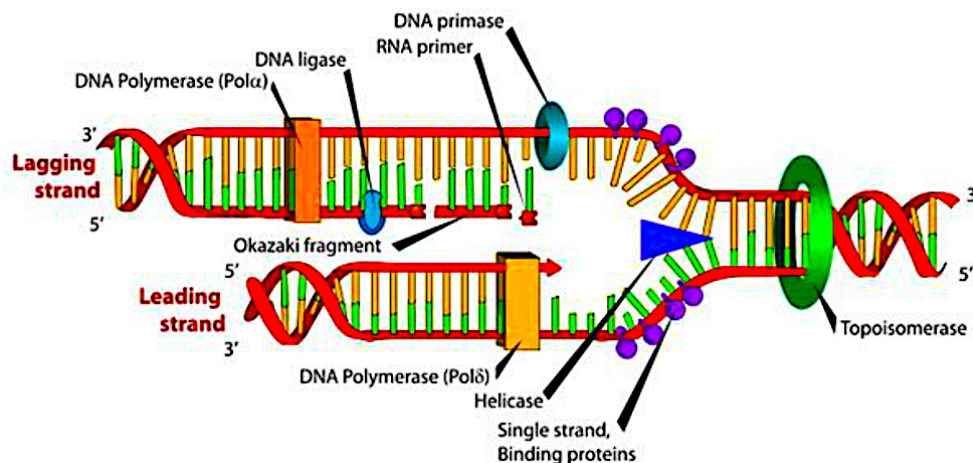


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HY Biochemistry

The purpose of this document is not to be a 600-page, long-winded textbook discussing every detail dating back to your high school bio days. The idea is to strike a balance between sufficient detail while staying as concise and HY as possible for USMLE (i.e., I'm aware some students want me to just cut to the chase with HY factoids; others want more in depth discussion of every mechanism). I will also make a separate doc specifically addressing more of the pedantic genetics stuff. Relax.

- Student question showed DNA being replicated and they pointed to enzyme at replication fork; what's the enzyme? → answer = helicase.
- Question asks what is acted upon by DNA ligase to form a DNA sequence that aligns complimentary with the leading strand in DNA synthesis? → answer = Okazaki fragments; the lagging strand in DNA synthesis is composed of Okazaki fragments that are synthesized discontinuously and then amalgamated using DNA ligase.



HY DNA replication enzymes for USMLE	
Enzyme	HY points
Topoisomerase	- Causes nicks in DNA; allows for relief of strand tension; prevents supercoiling and strand breakage. - Fluoroquinolones (e.g., levofloxacin, ciprofloxacin) are <i>prokaryotic</i> topoisomerase inhibitors (aka DNA gyrase). - Etoposide/teniposide → eukaryotic topoisomerase II inhibitors used as chemotherapeutic agents. - Irinotecan/topotecan → eukaryotic topoisomerase I inhibitors used as chemotherapeutic agents.
Primase	- Makes RNA primer at onset of DNA replication.
DNA polymerase III	- Synthesizes DNA 5' → 3' using the RNA primer synthesized by primase. - 3' → 5' exonuclease proofreading ability.
DNA polymerase I	- Degrades RNA primer after DNA polymerase III acts.
Telomerase	- Lengthens ends of DNA strands to prevent degradation; aging is correlated with loss of strand length; telomerase upregulated in some cancers (to prevent cell death).
Helicase	- Unwinds DNA template at the replication fork.
Ligase	- Creates a phosphodiesterase bond to join two nucleotides in growing DNA strand.

- 82M + prostatitis + treated with pharmacologic agent for 10 days + now has sore ankle while metal detecting; Q asks, what's the MOA of the drug he was treated with? → answer = inhibits DNA nicking; drug is fluoroquinolone (i.e., ciprofloxacin); frequent choice for prostatitis (and pyelonephritis); MOA is prokaryotic topoisomerase II/IV inhibitor (aka DNA gyrase); enzyme normally causes nicking in DNA to prevent supercoiling and breakage; HY adverse effect is tendonitis (e.g., Achilles); cannot be taken with divalent cations (i.e., foods containing calcium and iron → ↓ oral bioavailability of drug).
- 56M + lost in forest + eats mushrooms + dies; which of the following enzymes was most likely inhibited in this patient? → answer = RNA polymerase II; α -amanitin in death cap mushrooms inhibits RNA polymerase II (synthesizes mRNA).

RNA polymerases for USMLE		
Type	Target	HY point(s)
RNA polymerase I	rRNA	- Ribosomal RNA is a non-coding type of RNA that is the primary component of ribosomes. - Processing of rRNA into ribosomes occurs in nucleoli of the nucleus in eukaryotes and within the cytosol of prokaryotes (due to lack of nucleus).
RNA polymerase II	mRNA	- When DNA is transcribed into RNA, the RNA is first called hnRNA (heteronuclear RNA); three modifications then occur in the nucleus: 1) addition of 3' poly-A tail; 2) addition of 5' methylguanosine cap; 3) splicing out of introns; after these modifications, it is then called messenger RNA and exported to cytosol, where it is translated to protein. - mRNA can associate with p-bodies in the cytosol, which are "docking centers" where mRNA can be further processed and stored for future use. - RNA polymerase II + transcription factors bind to TATA box within the promoter sequence upstream of gene in order to commence transcription; if TATA box mutated → ↓ binding of RNA polymerase. - α-amanitin in death cap mushrooms inhibits RNA polymerase II.
RNA polymerase III	tRNA	- As the mRNA codon sequence is read on the ribosome, transfer-RNAs deliver amino acids and the growing peptide is synthesized. - tRNA anticodon sequence binds to complementary codon sequence of mRNA. - Different tRNAs carry different amino acids.

- 42F + has colonic polyps + tests positive for *MLH1* mutation + mom has Hx of duodenal and endometrial cancer; what's the most likely mechanism associated with this family's diagnosis? → answer = "microsatellite instability" or "defective mismatch repair"; Dx is Lynch syndrome (hereditary non-polyposis colorectal cancer; HNPCC); associated with mutations in mismatch repair genes *MLH1*, *MSH2*, *MSH6*, *PMS2*.
- Researcher is investigating a DNA repair process that is characterized by the deletion of an AGTC sequence within a DNA region rich in AGTC tandem repeats; he notices this DNA repair process is

inhibited when he adds DNase to the medium; Q asks which process this is → answer = slipped strand mispairing; student says wtf? → on the Step 1 NBME → sites rich in tandem sequences (e.g., AGTC) are prone to erroneous insertions/deletions of sequences; slipped strand mispairing can correct for these abnormalities; DNase disrupts process since the repair requires a transient breakage of the phosphodiester bond, where the DNA ends are exposed and can be degraded.

HY DNA repair types for USMLE	
Type	HY point(s)
Nucleotide-excision repair	- Removal of bulky helical distortions. - The answer if the Q asks about pyrimidine-pyrimidine dimers (i.e., from UV light). - Messed up in Xeroderma pigmentosum.
Base-excision repair	- Removal of single base. - The answer if the Q mentions glycosylases.
Mismatch repair	- Mismatched bases in newly synthesized (unmethylated) DNA strand are removed. - The answer if the Q says anything about “microsatellite instability”. - Messed up in Lynch syndrome (<i>HNPCC</i>). - There is a mutation process called slipped strand mispairing that is detected and corrected by mismatch repair. - Slipped strand mispairing (SSM) is the answer if the Q mentions an erroneous insertion or deletion of DNA repeat sequences within sites rich in tandem repeats (i.e., an extra AGTC is inserted within a site rich in AGTC repeats). The SSM process is disrupted in the presence of DNase.
Non-homologous end-joining	- Amalgamates ends of two dsDNA fragments. - The answer for / messed up in Ataxia-telangiectasia.
Recombinational dsDNA repair	- Defect in dsDNA is repaired using complementary strands from homologous dsDNA. - The answer if the Q asks about <i>BRCA</i> genes for breast cancer or Fanconi anemia.

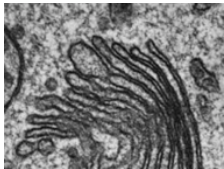
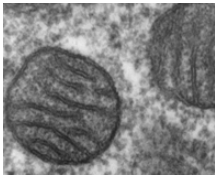
- 4M + congenital disorder characterized by mutation within intron sequence; which of the following processes is most likely to be disrupted in this patient? → answer = splicing; introns are sequences of DNA that do not eventually become protein; exons are sequences that can eventually be translated to protein; introns play a role in alternative splicing; mutations within intron sequences are known to sometimes cause problems with splicing.

HY DNA mutation types for USMLE	
Type	HY point(s)
Missense	- Codon sequence is altered, resulting in new codon coding for different amino acid – i.e., one amino acid is substituted for another. - The answer for sickle cell, where glutamic acid → valine in the beta-globin gene causes HbS.
Nonsense	- Codon sequence is altered such that a stop codon is formed, resulting in a truncated, non-functional protein. - Stop codons: UGA, UAA, UAG – “U Go Away, U Are Away, U Are Gone.” - (Start codon in eukaryotes is AUG, which codes for methionine).

	- The answer for Duchenne muscular dystrophy (can also be frame-shift mutation). Becker, in contrast, is usually caused by an in-frame deletion.
Silent	- Codon sequence is altered at third position, resulting in new codon that codes for the <i>same</i> amino acid – i.e., no change in protein primary sequence. - “Wobble” refers to the mutational “leniency” of this third codon position, where mutation does not cause change in the amino acid coded for. - “Wobble” is nature’s way of striking a balance between potentially advantageous vs deleterious mutations (i.e., allowing just enough mutations to enable evolution but not so much as to permit incessant pathology).
Frameshift	- Insertion/deletion of 1-2 bases, resulting in all following codons being read off-frame. - Effect is often truncated, non-functional protein, as stop codon may be erroneously formed. - The answer for Duchenne muscular dystrophy (can also be nonsense mutation). Becker, in contrast, is usually caused by an in-frame deletion.
Splice-site	- The answer if Q mentions a mutation within an intron sequence, or if the Q says an intron was erroneously retained within the final mRNA (i.e., it wasn’t spliced out). - β -thalassemia is HY condition often caused by splice-site mutations in intron sequences.

- Investigator is looking at a DNA locus many hundreds of bases away from a gene that causes upregulation of transcription of this gene; which of the following best describes this locus under investigation? → answer = enhancer; enhancers are DNA loci that may be near or far from a gene and cause upregulation of gene transcription; activators are proteins that bind to enhancers.
 - Silencers are the opposite of enhancers.
 - Repressors are proteins that bind silencer loci and cause transcriptional downregulation.
- Investigator is looking at prokaryotic translation process; which of the following sequences facilitates binding of bacterial ribosome to mRNA; answer = Shine-Dalgarno sequence; student says, “Yeah I’ve heard of that before. Wtf is that?” → ribosomal binding site on prokaryotic mRNA; aligns start codon on the mRNA with the ribosome to initiate translation.
 - Prokaryotic ribosomes: 30s + 50s subunits of 70s (correct, doesn’t add up)
 - 23s rRNA is component of 50S that is partial binding site for 50S inhibitors (i.e., macrolides, clindamycin, chloramphenicol, and linezolid); mutations in 23s RNA represent one mechanism of microbial resistance to these agents.
 - Eukaryotic ribosomes: 40s + 60s subunits of 80s (correct, doesn’t add up)
- What is a chaperone protein? → answer = facilitates protein folding (asked as one-liner on USMLE).
- 46M + pancreatic cancer + peripheral wasting; Q asks, which cellular organelle is most likely increased in this patient’s muscle cells? → NBME answer = autophagic vacuoles; autophagy is a process via which cellular components are broken down in cachexia and starvation; TNF- α is responsible for

cachexia in the setting of malignancy (aka cachectic factor); wrong answers are smooth endoplasmic reticulum, rough endoplasmic reticulum, Golgi, mitochondria, and mitotic spindles.

HY Cellular organelle points for USMLE	
Organelle	HY point(s)
Smooth endoplasmic reticulum (SER)	- Synthesizes steroids + some types of lipids. - Found notably in testes and adrenal glands (i.e., ↑ steroid biosynthesis).
Rough endoplasmic reticulum (RER)	- Synthesizes proteins mostly for exocytosis (i.e., for use outside the cell). - COP II transports proteins anterograde from RER to Golgi for processing / modification. - COP I transports proteins retrograde to RER from Golgi. - If Q tells you trafficking of proteins to Golgi is impaired, the answer for what intracellular change we expect is "dilation of RER" (makes sense, since proteins are "retained" at RER). - If you get a cystic fibrosis Q, the CFTR channel is misfolded and retained in the RER (i.e., do not choose misfolded channel on cell surface).
Free ribosomes	- Produce proteins that are used intracellularly (i.e., are not exocytosed).
Golgi 	- Processes and packages proteins for modification and exocytosis after receiving them from the RER. - The answer for I-cell disease if they ask for location of the enzyme defect. That is, the patient cannot synthesize mannose-6-phosphate at the Golgi in order to tag lysosomal proteins for trafficking to the lysosomes, so these lysosomal enzymes end up in the plasma instead.
Mitochondrion 	- Site of aerobic respiration (i.e., Krebs cycle + electron transport chain). - Drugs that cause uncoupling/diminution of proton gradient in intermembrane space lead to ↑ ratio of O₂ consumed per ATP produced (i.e., ↑ O₂ consumption/ATP production ratio → more O₂ needed to generate same amount of ATP → heat production). - Mitochondrial disorders present with buzzy triad of 1) ear/eye problems, 2) lactic acidosis, and 3) hypotonia. - Maternally inherited (i.e., males do not transmit disease to offspring). - "Ragged red fibers" is buzzy term for mitochondrial disorder on USMLE. - Heteroplasmy is varying severity of disease in patients with mitochondrial disorders due to differing fractions of diseased mitochondrial genes inherited. - For instance, a child might have decreased hearing and mild hypotonia, whereas his sister has diminished hearing + vision + severe lactic acidosis + severe hypotonia. - Any type of shock on USMLE (i.e., septic, cardiogenic, hypovolemic, etc.) can present with lactic acidosis due to ↓ oxygen delivery to tissues → ↓ ability for mitochondria to conduct aerobic respiration → ↑ anaerobic glycolysis, where pyruvate gets converted to lactate, rather than sent through the TCA cycle.
Centriole	- Produce mitotic spindles, which are composed of microtubules and form during M-phase of cell cycle (i.e., mitosis). - HY Drugs that inhibit microtubule formation: griseofulvin, mebendazole, colchicine, vincristine, vinblastine. - HY Drugs that inhibit microtubule disassembly: The taxanes (docetaxel, paclitaxel). These are the odd ones out because they hyperstabilize microtubules / prevent microtubule disassembly. The others prevent microtubule formation. - Griseofulvin → used orally for tinea capitis. Answer on USMLE is "oral griseofulvin for patient only." Giving to close contacts is wrong answer. - Mebendazole → used for many helminth infections.

	- Colchicine → Used for acute gout and pericarditis in patients who can't receive NSAIDs (i.e., GI bleeds, renal insufficiency, NSAID allergy). - Vincristine → used as chemotherapy; neurotoxic. Q will literally just give you massive paragraph where patient has neuro issues + is on vincristine, and the answer will be "toxic neuropathy." - Vinblastine → used as chemotherapy. - Taxanes → used as chemo. But just remember these are the odd ones out. - The USMLE can give you one of the above drugs and ask you what they inhibit, and the answer can be just "tubulin."
Nucleus	- Eukaryotes only. - Site of DNA replication and transcription to RNA. - Nucleoli are the site within the nucleus where rRNA is synthesized; "prominent nucleoli" in Q means ↑ protein synthesis and ↑ cancer grade (i.e., # of mitoses). - ↑ nuclear/cytoplasm ratio (e.g., if Q says it approaches 1:1) indicates high cancer <i>grade</i> – i.e., indicates ↑ mitoses. - Nuclear envelope proteins are phosphorylated during mitosis (ridiculous, but on Step 1 NBME).
Peroxisome	- Very long-chain and branched-chain fatty acid synthesis. - Refsum disease, adrenoleukodystrophy, Zellweger syndrome.
Lysosome	- Contain hydrolytic enzymes for digestive processes. - A phagolysosome refers to when a phagosome (the vesicle that is created during phagocytosis) merges with a lysosome. - Chediak-Higashi syndrome → phagolysosomal fusion disorder characterized by immunodeficiency, partial albinism, and giant granules within phagocytes. Melanosomes require phagolysosome function to be intact, which is why the Q can say the child has lighter skin than a sibling. And the "giant granules" descriptor is buzzy for Chediak-Higashi. - There are many lysosomal storage disorders, which I discuss later.
Proteasome	- Site of intracellular protein breakdown. - Ubiquitination of proteins targets them for lysosomes. This is extremely HY for USMLE. "Ubiquitin-mediated proteolysis" will be a frequent answer choice or stem variable on the exam. - For example, in the fed state (i.e., when insulin is high), ubiquitin-mediated proteolysis is ↓, since we'd have protein synthesis, not breakdown. - Likewise, in the fasting state, it is ↑, since we'd have protein catabolism. - In addition, if a patient experiences muscle atrophy, it is ↑, since we have protein breakdown. - Bortezomib is a proteasome inhibitor used for multiple myeloma that ↓ MHC I function. This results in ↓ CD8+ T cell activity by extension, since the latter normally interacts with MHC I. This is asked on the NBME. Not writing about this garbage because I find it entertaining.

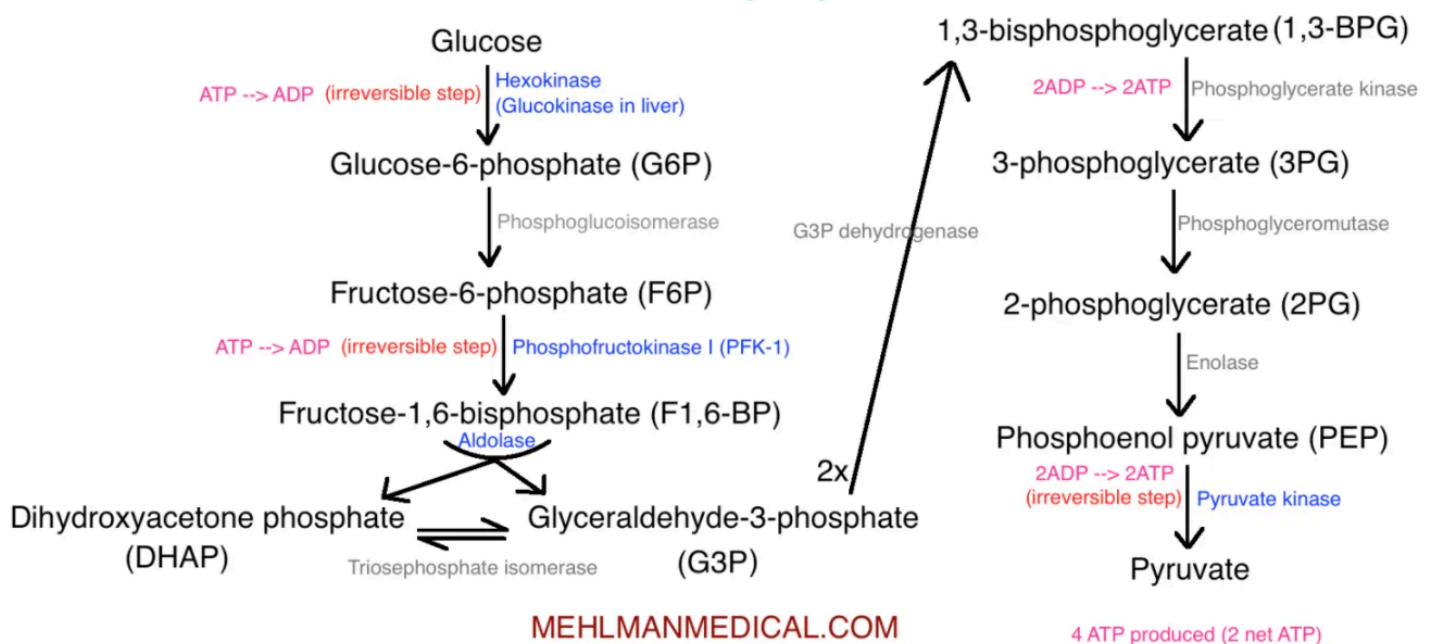
- 34F + undergoing chemotherapy; which of the following cell types in this patient is most likely to be adversely impacted by this management? → answer = crypts of gut epithelium (any answer that is a labile cell characterized by rapid turnover); crypts are site of gut stem cells.

Labile, Stable, Permanent cells	
Cell type	HY points
Labile	- Remain in G1; never enter G0 (quiescent phase). - HY examples: Gut epithelium, hair follicles, skin, germ cells, bone marrow. - Most affected by chemo-/radiotherapy because of rapid turnover.

	- USMLE will tell you someone is exposed to radiation then ask for what they're most likely to experience in the next 24 hours → answer = gastrointestinal disturbance, where the other answers listed aren't labile.
Stable (quiescent)	- Enter G1 from G0 when stimulated. - HY Examples: Hepatocytes, renal tubular cells. - Regeneration of liver → answer = "enter G1 from G0". - USMLE will mention liver regenerative ability, and then the answer is something like, "reemergence from G0 quiescent phase."
Permanent	- Always in G0; regenerate from stem cells. - HY examples: Neurons, cardiac + skeletal muscle cells. - USMLE will tell you someone is exposed to radiation + ask what is most likely to be unaffected, answer is "cardiac myocytes" (the only answer listed that isn't labile).

- 23M + consumes energy drink; which of the following irreversible enzymes is most likely to demonstrate increased activity? → answer = pyruvate kinase (of glycolysis).
 - o An "easy" way to remember the irreversible enzymes in glycolysis is that they are enzyme #s 1, 3, 9 – i.e., hexokinase/glucokinase (#1), PFK-1 (#3), pyruvate kinase (#9).

Glycolysis

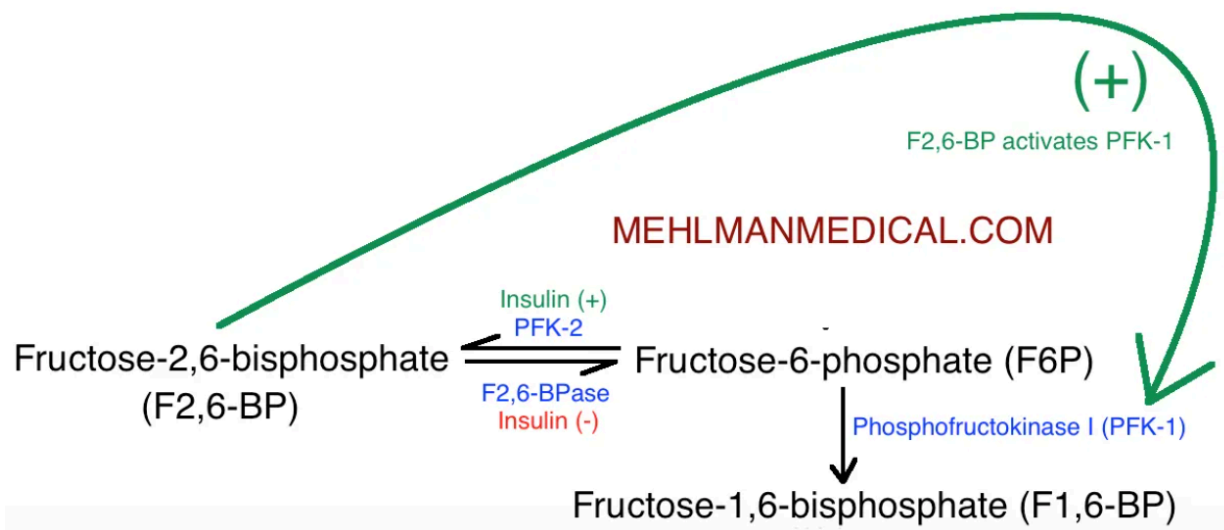


- 17F + frequent thirst and urination + markedly elevated serum glucose and ketones + serum pH of 7.1; following administration of insulin; activity of which enzyme is increased? → answer = glucokinase; glucokinase is the hexokinase equivalent in the liver and is upregulated by insulin; hexokinase is found elsewhere in the body and not upregulated by insulin; compared to hexokinase, glucokinase has ↑ K_m and ↑ V_{max} , meaning that it has less affinity for glucose and greater capacity to

handle it; this makes sense, since the liver should not preferentially process glucose over other cells in the body (e.g., muscle, brain); when serum glucose levels have risen high enough, the liver can act as a buffer to process glucose into glycogen (glycogenesis).

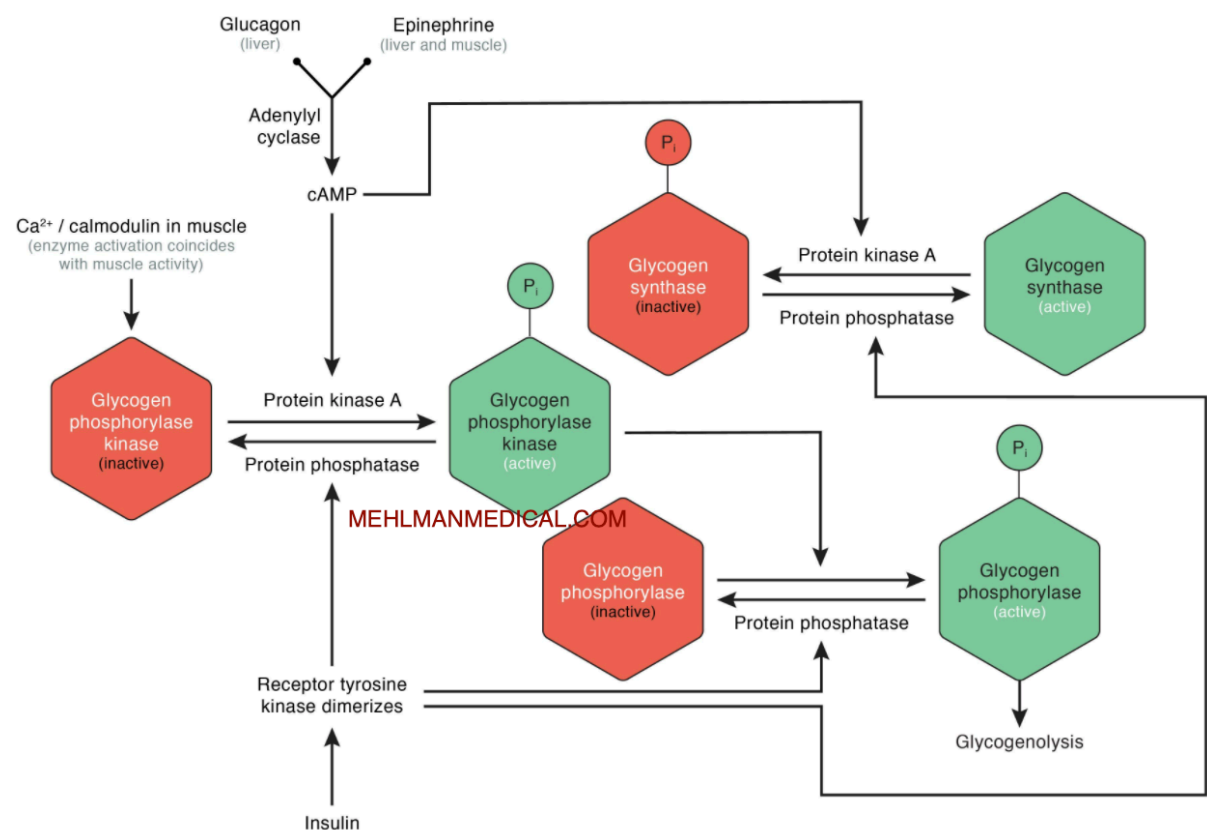
Enzyme	Where	Km	Vmax	Upregulated by insulin?
Hexokinase	Mostly ubiquitous	↓	↓	No
Glucokinase	Liver + β -islets	↑	↑	Yes

- 19F + eats meal; which of the following molecules is most likely to stimulate glycolysis in this patient; answer = AMP; AMP + ADP stimulate glycolysis; ATP, NADH, citrate, and alanine inhibit glycolysis.
- 61F + eats meal; which of the following molecules is most likely to upregulate the rate-limiting step of glucose utilization in this patient; answer = fructose-2,6-bisphosphate.



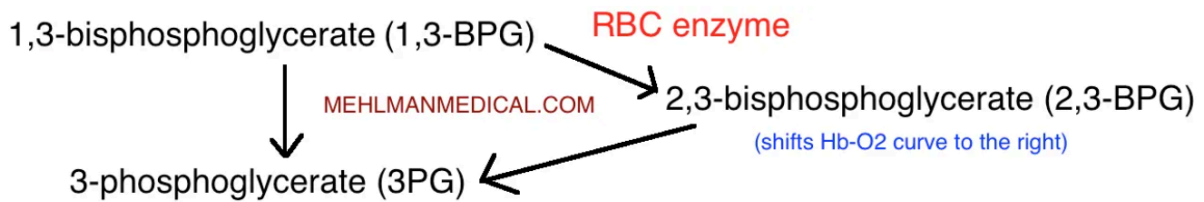
- PFK-1 is the rate-limiting step of glycolysis. F2,6-BP is a positive allosteric regulator of PFK-1. When insulin is high (fed state), some of F6P from glycolysis, rather than simply proceeding forward through glycolysis to F1,6-BP, is instead shunted into a side pathway to make F2,6-BP, which then positively allosterically regulates PFK-1.
- 47F + goes for a run; which of the following combinations best reflects glycogen phosphorylase in this patient (answers are combinations of phosphorylated vs dephosphorylated, active vs inactive); answer = phosphorylated + active. Glycogen phosphorylase breaks down glycogen (fasting state). Glycogen synthase builds up glycogen (fed state). Insulin dephosphorylates. Glucagon phosphorylates.
 - Fed state ($\uparrow$ insulin): enzymes dephosphorylated.
 - Fasting state ($\downarrow$ insulin): enzymes phosphorylated.

- 1st step: “Are we fed or fasting?”
 - 2nd step: “Would we expect a given enzyme to be active or inactive in this setting?”
 - 3rd step: “Is insulin up or down?”
 - 4th step: If insulin is up, then enzyme is dephosphorylated. If insulin is down, then enzyme is phosphorylated.
- 27F + eats turkey dinner; which of the following combinations, in terms of inactive vs active, phosphorylated vs dephosphorylated, best reflects her phosphofructokinase-2? → answer = active + dephosphorylated; scenario is fed state ($\uparrow$ insulin), enzyme upregulates glycolysis, so we expect it to be active in fed state. Since insulin is up, we choose dephosphorylated.



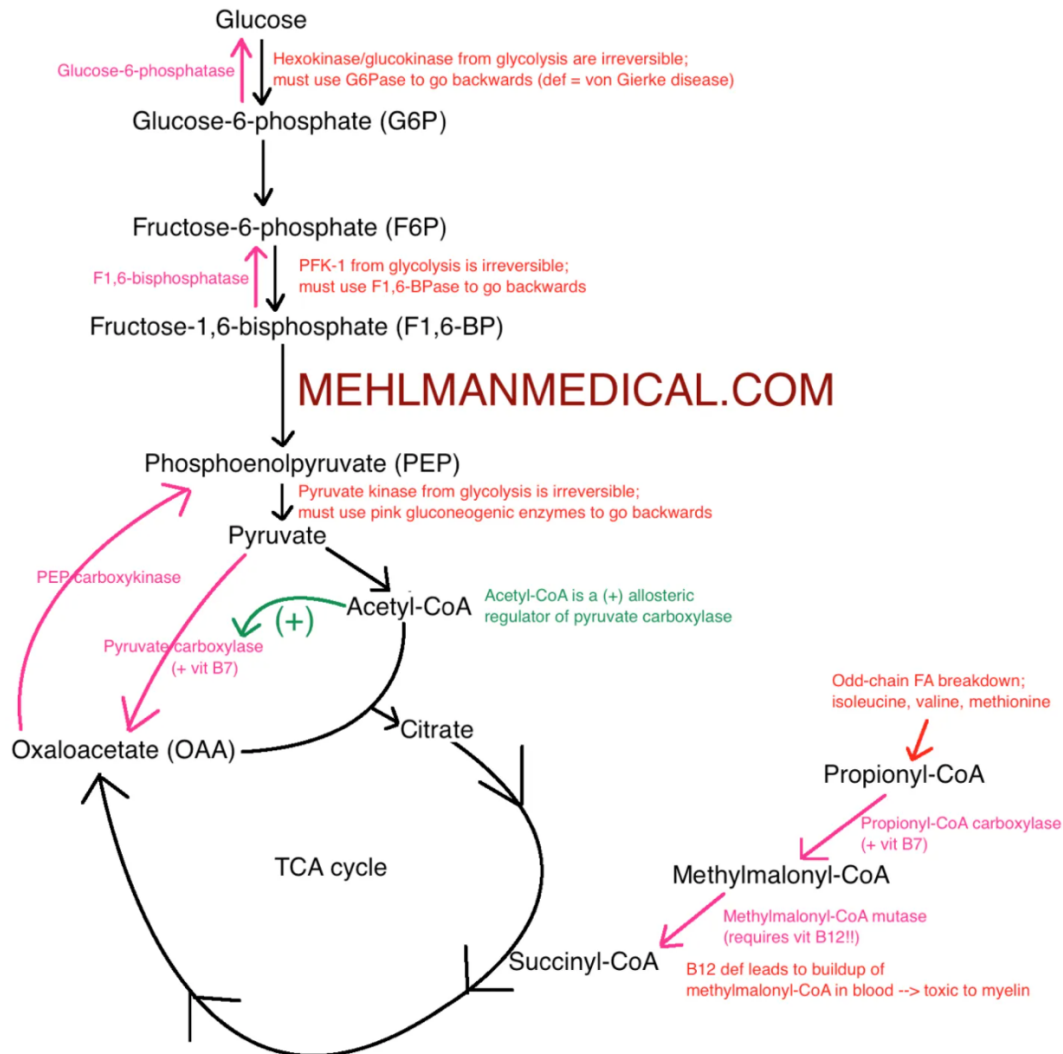
- 36F + type I diabetic + self-administers insulin; which of the following best explains the reduction of serum glucose in this patient? → answer = dephosphorylation of glycogen synthase; insulin upregulates GLUT4 on adipose tissue and skeletal muscle, yes, but it also upregulates glucokinase in the liver → liver acts as a buffer in the setting of increased serum glucose → glucose converted over to glycogen secondary to the dephosphorylation and activation of glycogen synthase by insulin.
- Student says “Oh em gee. I thought it was just GLUT4.” Yeah, I know. Incredible.

- Research group is investigating a protein in RBCs that shifts the Hb-O₂ dissociation curve to the right. It is concluded that this molecule can be synthesized from a glycolytic intermediate via an RBC enzyme called bisphosphoglycerate mutase; which of the following glycolytic substrates is most likely the precursor to the RBC protein? → answer = 1,3-bisphosphoglycerate (1,3-BPG). The RBC protein is 2,3-BPG. Some 1,3-BPG from glycolysis is converted to 2,3-BPG inside the RBC via bisphosphoglycerate mutase.



- 20M + low Hb + high reticulocyte count + high indirect bilirubin + high RBC 2,3-BPG; which enzyme is deficient? → answer = pyruvate kinase deficiency; second most common cause of hemolysis due to an enzyme deficiency (after G6PD deficiency); pyruvate kinase converts phosphoenol pyruvate → pyruvate as last step of glycolysis; if decreased production of pyruvate, then decreased ATP production → decreased activity of Na/K-ATPase pumps on RBC → cannot pump out sodium → cellular swelling + lysis (hemolysis); wrong answer is glyceraldehyde-3-phosphate dehydrogenase deficiency (would mean decreased 1,3-BPG, and ultimately decreased 2,3-BPG).
- 21M + goes three weeks without food; breakdown of which of the following most likely explains his ability to maintain stable blood glucose levels? → answer = skeletal muscle protein; glucogenic amino acids will be liberated by muscle → converted to glucose by the liver; muscle lacks glucose-6-phosphatase so does not directly produce glucose via gluconeogenesis.
- 22F + goes for a run; Q asks which enzyme will most likely be activated initially by exercise in this patient; answer = phosphorylase kinase, which will phosphorylate and activate glycogen phosphorylase, the main enzyme that removes glucose-1-phosphate (G1P) units from long glycogen branches and chains. In other words, before glycogen phosphorylase can do its job, it first needs to be phosphorylated and activated by phosphorylase kinase. After the glycogen branch has enough G1P units removed where it is now only four G1P units in length, the first debranching enzyme 1,4-glucosidase removes three of these units, leaving only one left. The second debranching enzyme 1,6-glucosidase removes the final G1P unit.

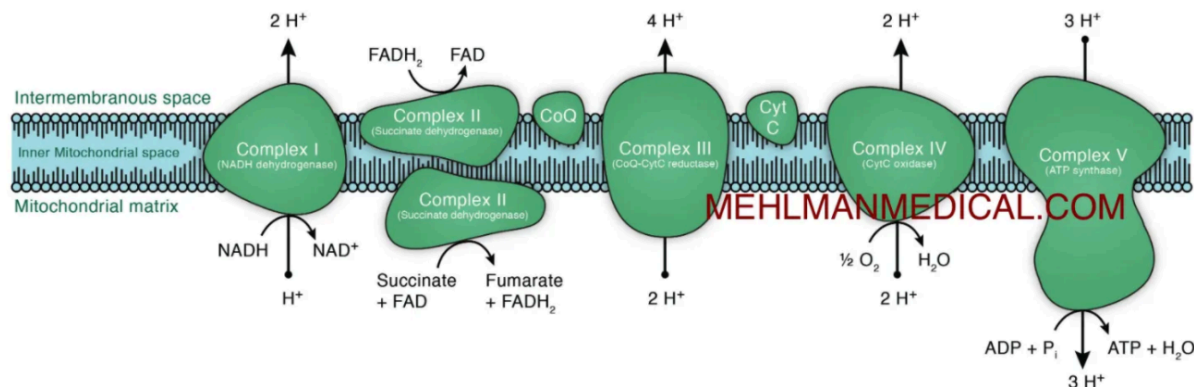
GLUCONEOGENESIS



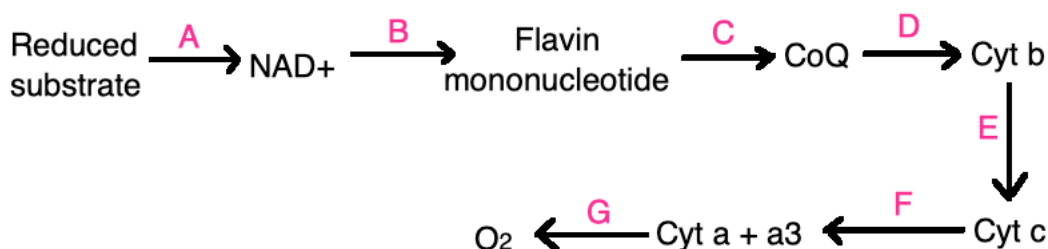
- 10-month-old boy + 3rd percentile for length and weight + hypotonia + hypoglycemia + lactic acidosis + hyperalaninemia; Q asks which enzyme is deficient; answer = pyruvate carboxylase; enzyme needed to convert pyruvate to oxaloacetate (OAA) as an early step for gluconeogenesis (so ↓ glucose); if less pyruvate is converted to OAA, then more pyruvate is shunted to alanine (pyruvate + glutamate $\leftrightarrow$ α -KG + alanine, via ALT and B6); more pyruvate is also converted to lactate via lactate dehydrogenase.
- Experiment performed with brown adipose tissue; discovery shows a leak of H⁺ ions inward across the inner mitochondrial membrane; Q asks, most likely effect on oxidative phosphorylation and energy metabolism? → answer = "increased ratio of oxygen consumption to ATP generated"; normally H⁺ in intermembrane space moves through Complex V (ATP synthase) in order to produce ATP, but if H⁺ leaks back across the inner membrane without going through Complex V (i.e., as a result of an

uncoupling agent such as thermogenin in brown fat), more oxygen is needed to achieve the same # of ATP produced → generates heat.

Electron transport chain

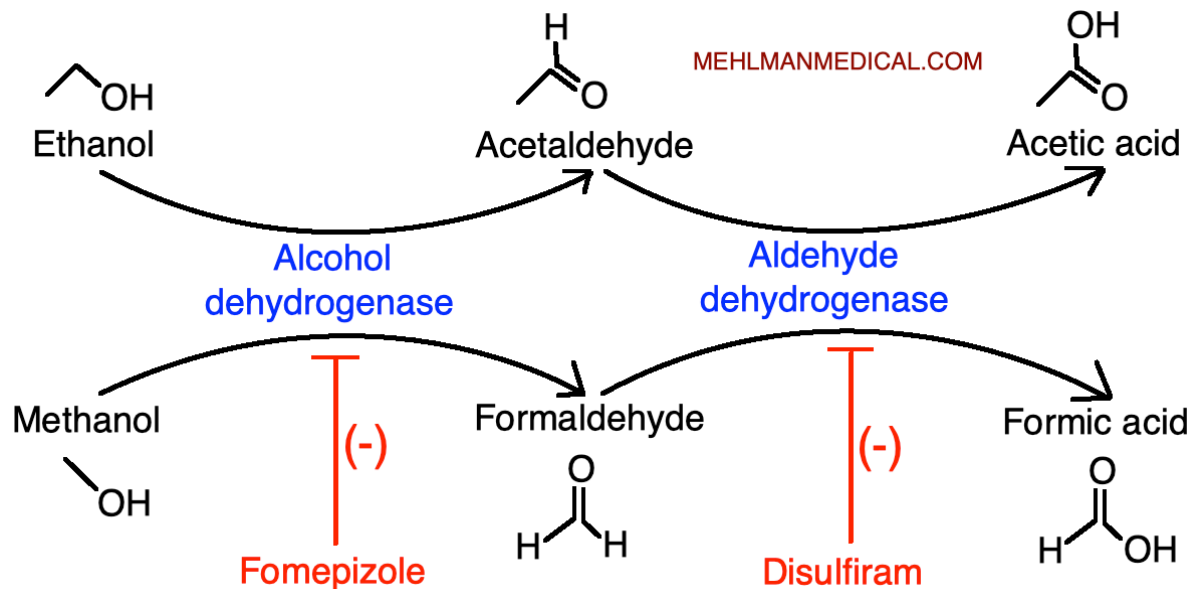


- 45M + wood worker + exposed to preservative for the wood that increases heat production in his cells; which of the following mechanisms best describes this process? → answer = “uncoupling of oxidative phosphorylation”; 2,4-dinitrophenol is a preservative for wood that is best known for dissipating the proton gradient required for oxidative phosphorylation (electron transport chain).
 - Uncoupling agents include 2,4-Dinitrophenol, Ethanol, Aspirin, Thermogenin (DEATH).
 - Rotenone inhibits Complex I (NADH dehydrogenase).
 - Antimycin A inhibits Complex III.
 - Carbon monoxide (CO) inhibits Cytochrome a3 component of Complex IV (cytochrome c oxidase).
- 2M + diminished mental status + family recently moved to winter lodge with old ventilator + carboxyhemoglobin elevated; which of the following steps is inhibited in this patient?

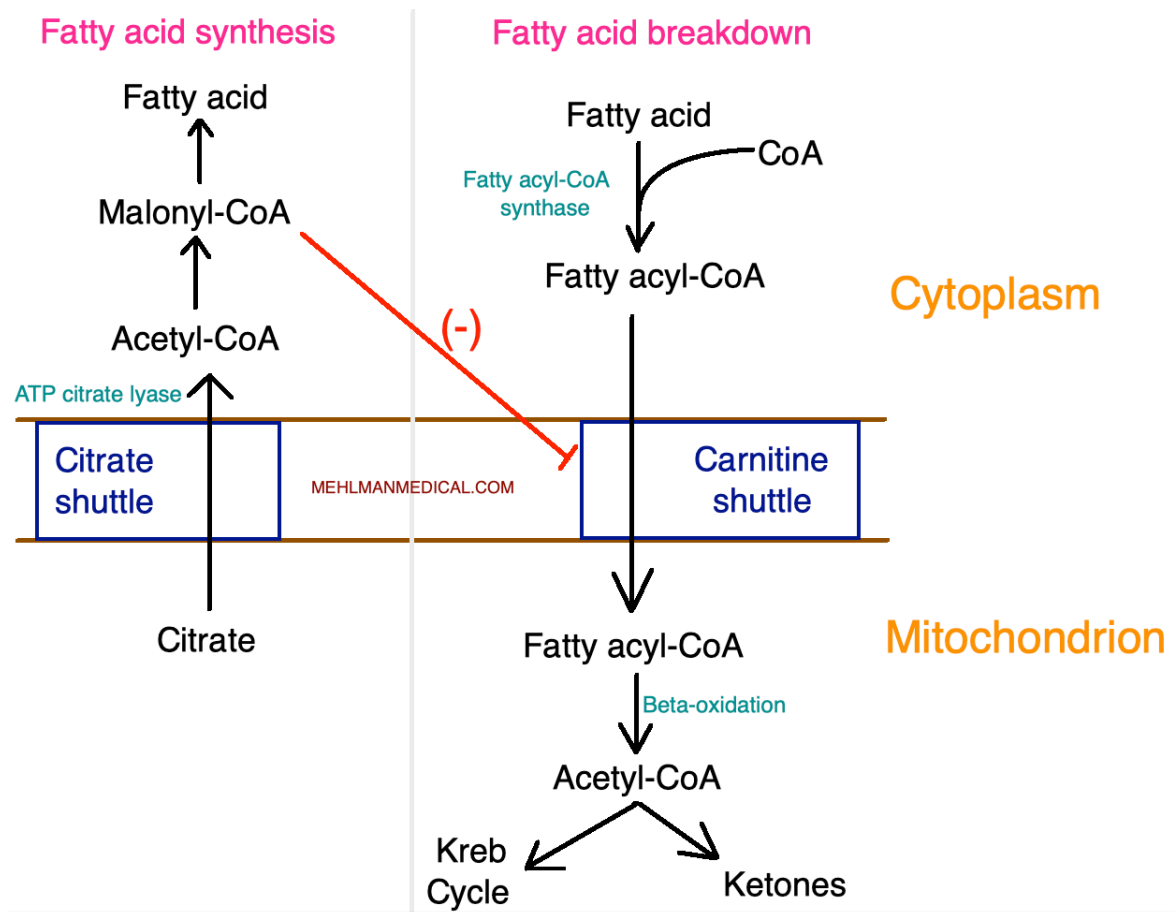


- Answer = G; CO inhibits Cyt a3 component of Complex IV (cytochrome c oxidase). Yes, I know. Outrageous. But it's not our opinion that matters. It's on a retired Step 1 NBME, so just know it.

- 40M + working with paint thinner + vomiting + blurry vision + serum pH 7.27 + high-anion gap; Q asks for the pharmacologic treatment? → answer = fomepizole; diagnosis is methanol toxicity; methanol is in paint thinner; Alcohol dehydrogenase converts methanol to formaldehyde, which can cause blindness and death; fomepizole inhibits this conversion by inhibiting alcohol dehydrogenase.



- Neonate + has defect in fatty acid oxidation + physical exam shows no abnormalities; Q asks, what's the next best step in diagnosis? → answer = "check serum acylcarnitine concentrations"; acylcarnitines are produced during the movement of fatty acids from the cytosol to the mitochondria via the carnitine shuttle; low acylcarnitines suggests carnitine deficiency; normal acylcarnitines suggests medium- or long-chain acylCoA dehydrogenase (MCAD/LCAD) deficiency; affected children will have **hypoketotic hypoglycemia** (i.e., low ketones and low glucose), since ketones cannot be produced without effective beta-oxidation; ketones are produced via the assembly of acetyl-CoA units; the latter are liberated during the breakdown of fatty acids via beta-oxidation. In other words: ↓ beta-oxidation → ↓ acetyl-CoA → ↓ ketogenesis. And since acetyl-CoA is a positive allosteric regulator of pyruvate carboxylase (enzyme used in gluconeogenesis, as discussed earlier), if acetyl-CoA is ↓, then glucose production is ↓.



- Researcher is conducting study on pyruvate carboxylase; which of the following molecules is found to increase activity of this enzyme? → answer = acetyl-CoA.
- 2M + hepatosplenomegaly + Hx of cardiopulmonary arrest following hypoglycemic episode + ↓ serum glucose + ↓ serum ketones + administration of medium-chain triglycerides improves his condition; diagnosis? → answer = long-chain acyl-CoA dehydrogenase (LCAD) deficiency; patient has hypoketotic hypoglycemia, so the answer you want to look for is either carnitine, MCAD, or LCAD deficiency; since medium-chain TGAs improved his condition, you know it can't be MCAD deficiency (because he wouldn't be able to process them), so LCAD deficiency is correct; for carnitine deficiency, neither medium- or long-chain TGAs would improve the condition because the carnitine shuttle would be defective (cannot move fatty acids from cytosol to mitochondria).
- 3M + disorder characterized by inability of epinephrine to liberate fatty acids for energy; Q asks, which hormone is likely deficient in this patient? → answer = hormone-sensitive lipase (HSL); HSL is catabolic and required to move TGAs from the adipocyte into the blood; it has increased activity in the setting of higher levels of epinephrine, cortisol, and/or glucagon. In contrast, lipoprotein lipase

(LPL) is anabolic and moves TGAs from the blood into the adipocyte; it has increased activity with higher levels of insulin. Other HY points: deficiency of LPL or apolipoprotein C-II causes familial hyperchylomicronemia ($\uparrow$ TGAs and LDL); fibrates (e.g., fenofibrate, gemfibrozil) upregulate LPL activity and are most effective at $\downarrow$ serum TGAs.

- 3M + acanthocytes seen on blood smear + intestinal biopsy shows large clear droplets within the enterocytes; Dx? $\rightarrow$ answer = abetalipoproteinemia; $\downarrow$ apolipoprotein-B48 $\rightarrow$ chylomicrons fail to exit enterocytes.

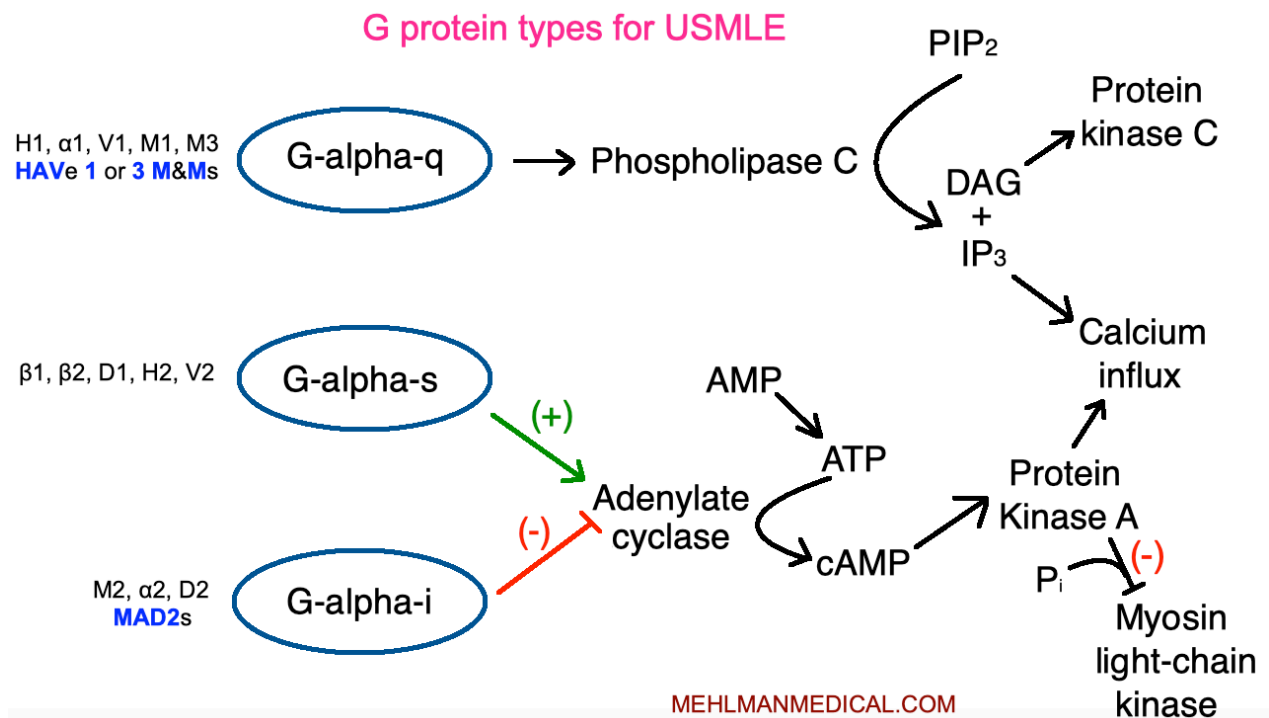
Apolipoprotein HY points for USMLE	
Apolipoprotein	HY points
B-48	- Required to move chylomicrons out of the enterocytes into lymphatics/circulation. - Deficient in abetalipoproteinemia (acanthocytes; clear droplets in enterocytes).
B-100	- Binds to LDL receptor; enables clearance of cholesterol from blood.
C-II	- Cofactor for lipoprotein lipase (LPL). - Deficient in familial hyperchylomicronemia (or deficiency of LPL). - Fibrates upregulate lipoprotein lipase and Apo-CII and are most effective med for $\downarrow$ TGAs.
E	- Mediates particle uptake by the liver.
A	- Major protein component of HDL.

- 14F + LDL of 280 mg/dL + father died of MI in his early-40s; what's the mechanism for this patient's condition? $\rightarrow$ answer = deficiency of LDL receptor; Dx is familial hypercholesterolemia.

HY familial dyslipidemias for USMLE	
Condition	HY point(s)
Hyperchylomicronemia (AR)	- Deficiency of lipoprotein lipase or Apolipoprotein C-II. - $\uparrow$ chylomicrons + TGAs. - The answer on USMLE if the Q says the plasma appears "creamy." - Pancreatitis (abdo pain more likely than chest pain) and xanthomas.
Hypercholesterolemia (AD)	- Deficiency (heterozygous) or absence (homozygous) of LDL receptor or Apo B-100. - $\uparrow$ LDL. Usually 3-400 in heterozygotes, with MI/death in 30-40s (answer = deficiency of functional LDL receptor). - LDL usually 700-1000 in homozygotes, with MI in teens (answer = absence of functional LDL receptor). - The answer for early death from MI that is familial. Can cause Xanthomas.
Hypertriglyceridemia (AD)	- $\uparrow$ Hepatic production of VLDL. - $\uparrow$ TGAs on lab report. LDL will be normal. - Pancreatitis.

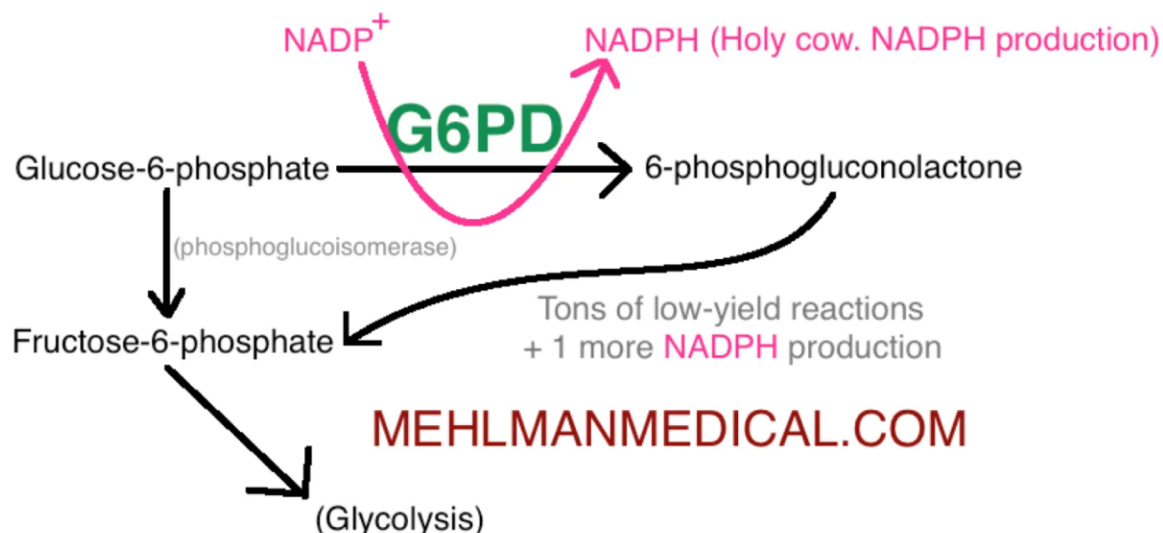
- Investigator is studying vancomycin-resistant bacterial strain; after many generations of propagation, vancomycin-sensitivity is observed at a frequency of one per 200 cells; what's the mechanism for loss of resistance? $\rightarrow$ answer = "plasmid loss"; most antibiotics resistance genes are located on the bacterial *plasmid*.

- Plasmids have three main components: 1) origin of replication (where DNA replication is initiated), 2) polylinker (site where endonucleases act), 3) antibiotic resistance gene.
- Cloning is process of replicating ("cloning") genetic material up to 10k base pairs. Process is:
 - Cleave out polylinker + insert desired gene of interest to be cloned (i.e., the gene you want to clone replaces the polylinker you've cleaved out).
 - Incubate plasmid with bacteria → bacteria then acquire the plasmid containing both the desired gene to be cloned and the antibiotic resistance gene.
 - Incubate bacteria with antibiotic → only bacteria replicating the plasmid survive.
 - Extract out the cloned/desired DNA/protein from bacteria cells containing plasmid.
- Researcher develops toxin that causes inhibition of GTPase activity of G-alpha-s G proteins; which would most likely increase in a cell as a result of this toxin? → answer = cAMP; GTPases normally function to shut off G-protein function; if GTPase activity is impaired (e.g., in cancer), there is constitutive (continual; increased) activity of the G-protein.
- G-proteins are incredibly annoying concept if you're seeing this concept for first time, but you need to know three classes for USMLE and how agonism/antagonism causes intracellular changes:

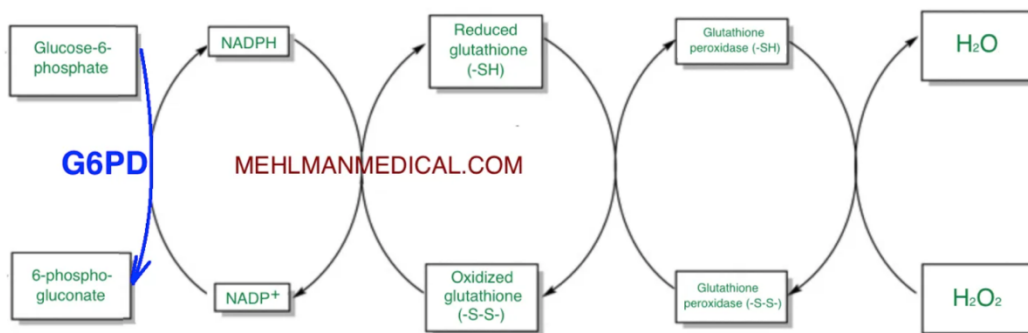
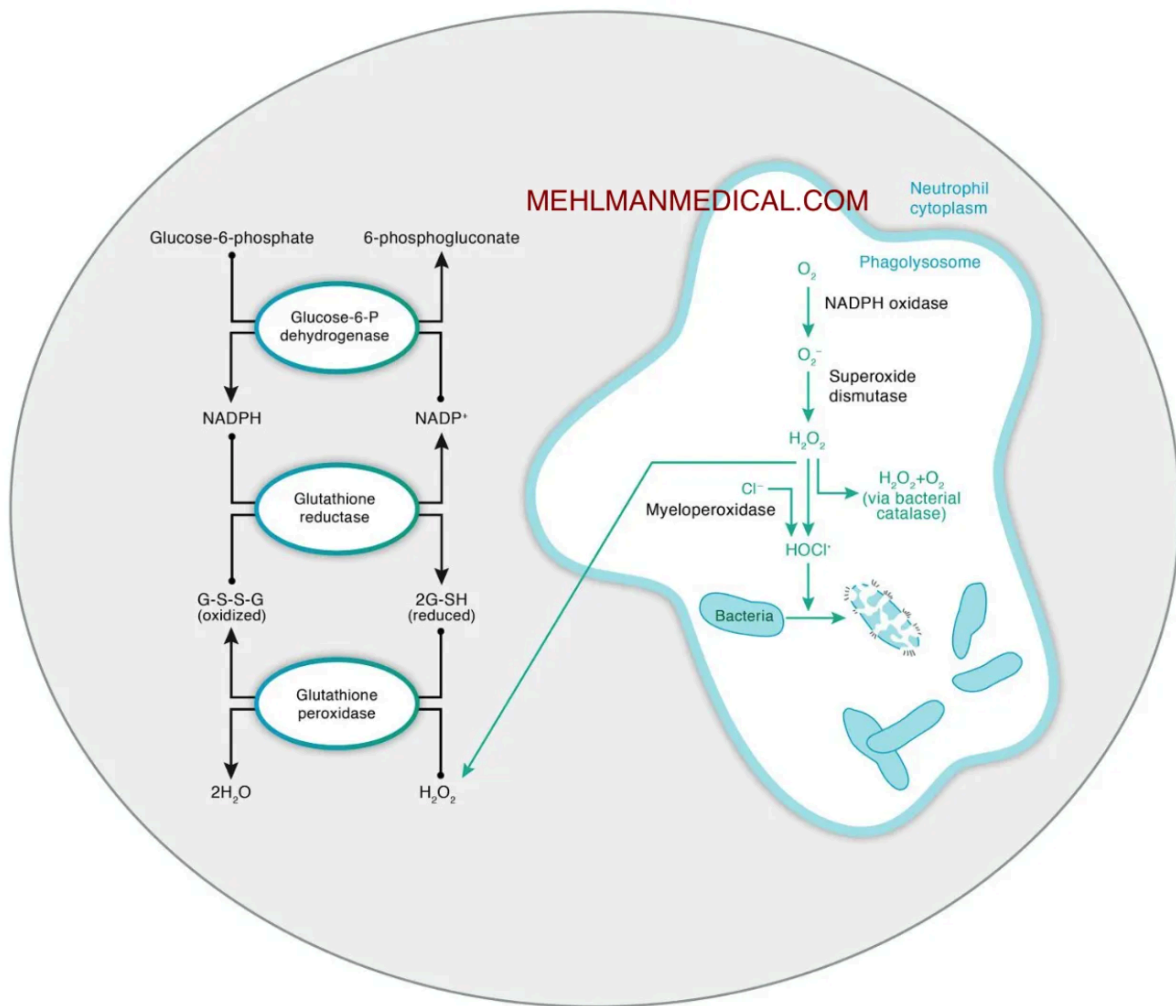


- Agonism of G-alpha-q G proteins → increases phospholipase C, inositol 1,4,5-triphosphate (IP3), and diacylglycerol (yes, you need to know all three).
 - Agonism of G-alpha-i → inhibits adenylyl cyclase → decreases cAMP.
 - Agonism of G-alpha-s → activates adenylyl cyclase → increases cAMP.
 - Receptors for G-alpha-q: **HAVE 1** or **3 M&Ms** → Histamine 1, Alpha 1, Vasopressin 1, Muscarinic 1, Muscarinic 3.
 - In other words, if you give phenylephrine (alpha 1 agonist), you're agonizing G-alpha-q, so the answer would be "increased IP3," or "increased diacylglycerol."
 - Likewise, if you give diphenhydramine (H1 blocker), the answer would be "decreased IP3," or "decreased phospholipase C activity."
 - Receptors for G-alpha-i: **MAD2s** → Muscarinic 2, Alpha 2, Dopamine 2.
 - In other words, if you give methyldopa or clonidine (alpha 2 agonists), the answer is "decreased cAMP."
 - Likewise, if you give haloperidol (D2 blocker), the answer is "increased cAMP."
 - Receptors for G-alpha-s: Beta 1, Beta 2, Dopamine 1, Histamine 2, Vasopressin 2 (essentially memorize both aforementioned mnemonics, and then the receptors that don't fit into either you know are G-alpha-s).
 - In other words, for isoproterenol (beta 1 and 2 agonist), the answer is "increased cAMP."
 - Likewise, for propranolol (beta 1 and 2 antagonist), the answer is "decreased cAMP."
- Researcher studying influenza virus and antigenic shift; which of the following mechanisms best explains this process? → answer = "reassortment"; antigenic shift → pandemics → reassortment of viral segments; antigenic drift → epidemics → point mutations in hemagglutinin and/or neuraminidase.
- 13M + yellow eyes following treatment for infection + blood smear shows precipitated hemoglobin within RBCs; what's the most likely mechanism for this condition? → "increased RBC membrane oxidation"; Dx is glucose-6-phosphate dehydrogenase (G6PD) deficiency; X-linked recessive; due to

deficient NADPH production via the HMP shunt; NADPH is a reducing agent that ↓ oxidation of RBC membranes; patients have ↑ hemolysis with certain drugs (i.e., sulfa, dapsone, hydroxychloroquine), infections, and, famously, fava beans. Blood smear shows degmacytes (bite cells) and Heinz bodies (precipitated, oxidized hemoglobin).



- 19F + enzyme deficiency + increased indirect bilirubin; which enzyme is most likely deficient in this patient? → answer = pyruvate kinase deficiency → RBC cannot make sufficient ATP to drive Na out of the cell → water stays with sodium → RBC swelling → lysis; second most common cause of hemolysis due to an enzyme deficiency (after G6PD deficiency); pyruvate kinase deficiency is AR so can occur in females; G6PD (XR) only presents in males.
- 12M + Hx of recurrent Staph infections + dihydrorhodamine test confirms diagnosis; the deficient enzyme in this patient acts on which of the following as a substrate? → answer = molecular oxygen; diagnosis is NADPH oxidase deficiency (chronic granulomatous disease); nitroblue tetrazolium assay now obsolete and is the wrong answer on one of the Step 1 NBMEs (dihydrorhodamine test being correct); NADPH oxidase required as initial enzyme in order to ultimately generate sufficient H_2O_2 to overwhelm catalase (+) organisms; even though the latter generate catalase, which breaks down H_2O_2 , healthy individuals produce enough H_2O_2 via respiratory burst pathway to overpower these pathogens. Molecular oxygen (O_2) is the substrate of NADPH oxidase; should be noted that myeloperoxidase deficiency leading to insufficient generation of halide-hydroxyl radicals (bleach; hypochlorous acid) is equally HY on NBME yet underemphasized in other resources. NADPH is a reducing agent used in a variety of reactions produced via the G6PD pathway (HMP shunt).



- 6-month-old boy + enlarged tongue + hypotonia + cardiomegaly + muscle biopsy shows increased glycogen; Q asks which enzyme is deficient → answer = alpha-1-4-glucosidase (aka lysosomal acid maltase); diagnosis is Pompe syndrome (glycogen storage disease type II).
- 4-month-old boy + 2-month Hx of frequent crying and tremulousness presenting a few hours after feeds + massive hepatomegaly + hypoglycemia + lactic acidosis; which enzyme is deficient? → answer

= glucose-6-phosphatase; Dx is von Gierke (glycogen storage disease type I); classically presents as young child with hepato-/hepatosplenomegaly, lactic acidosis, and hypoglycemia.

- **Glycogen storage diseases**

- All autosomal recessive and can present with hepatomegaly.

Glycogen storage diseases	
Disease	HY points
Type I (Von Gierke)	- Deficiency of glucose-6-phosphatase. - Lactic acidosis. - Hypoglycemia, jaundice, hepatomegaly. - "Super sick kid with glycogen storage disease."
Type II (Pompe)	- Deficiency of α1,4-glucosidase (aka lysosomal acid maltase). - Cardiomyopathy, hepatomegaly.
Type III (Cori)	- Deficiency of α1,6-glucosidase (aka "debranching enzyme"). - No lactic acidosis (in contrast to von Gierke). - "Not so sick kid with glycogen storage disease."
Type V (McArdle)	- Deficiency of myophosphorylase (glycogen muscle phosphorylase). - Patient usually adolescent or adult. - Severe cramping/rhabdo after intense exercise. - No lactic acidosis.

- **Lysosomal storage diseases**

- All AR except for Fabry and Hunter (XR).
- 4M + coarse facial features + contractures of large joints + markedly elevated plasma lysosomal enzyme levels; Q asks, what's the most likely mechanism for this condition? → answer = "abnormal targeting of enzymes to lysosomes"; diagnosis is I-cell disease; as discussed, mechanism is inability of the Golgi to make mannose-6-phosphate, which is required for proper targeting of lysosomal hydrolases to the lysosomes for storage; if they can't go to the lysosomes, they are secreted into the plasma.

Lysosomal storage diseases			
Disease	Enzyme deficiency	Buildup product(s)	HY points
I-cell	N-acetylglucosamine-1-phosphotransferase	Many lysosomal enzymes	- Cannot synthesize mannose-6-phosphate at the Golgi; M6P needed to target lysosomal enzymes to the lysosomes; instead, lysosomal enzymes are secreted into cytosol + serum. - Coarse facial features, joint contractures, hepatomegaly, recurrent ear infections. - If the USMLE Q asks for the location of the error/problem, Golgi is the answer. - If the Q asks for where the enzymes <i>can't go</i>, the answer is lysosome. - If the Q asks for where the enzymes <i>do go</i>, the answer is cytosol/plasma/serum.
Gaucher	Glucocerebrosidase	Glucocerebroside	- Avascular necrosis of the hip.

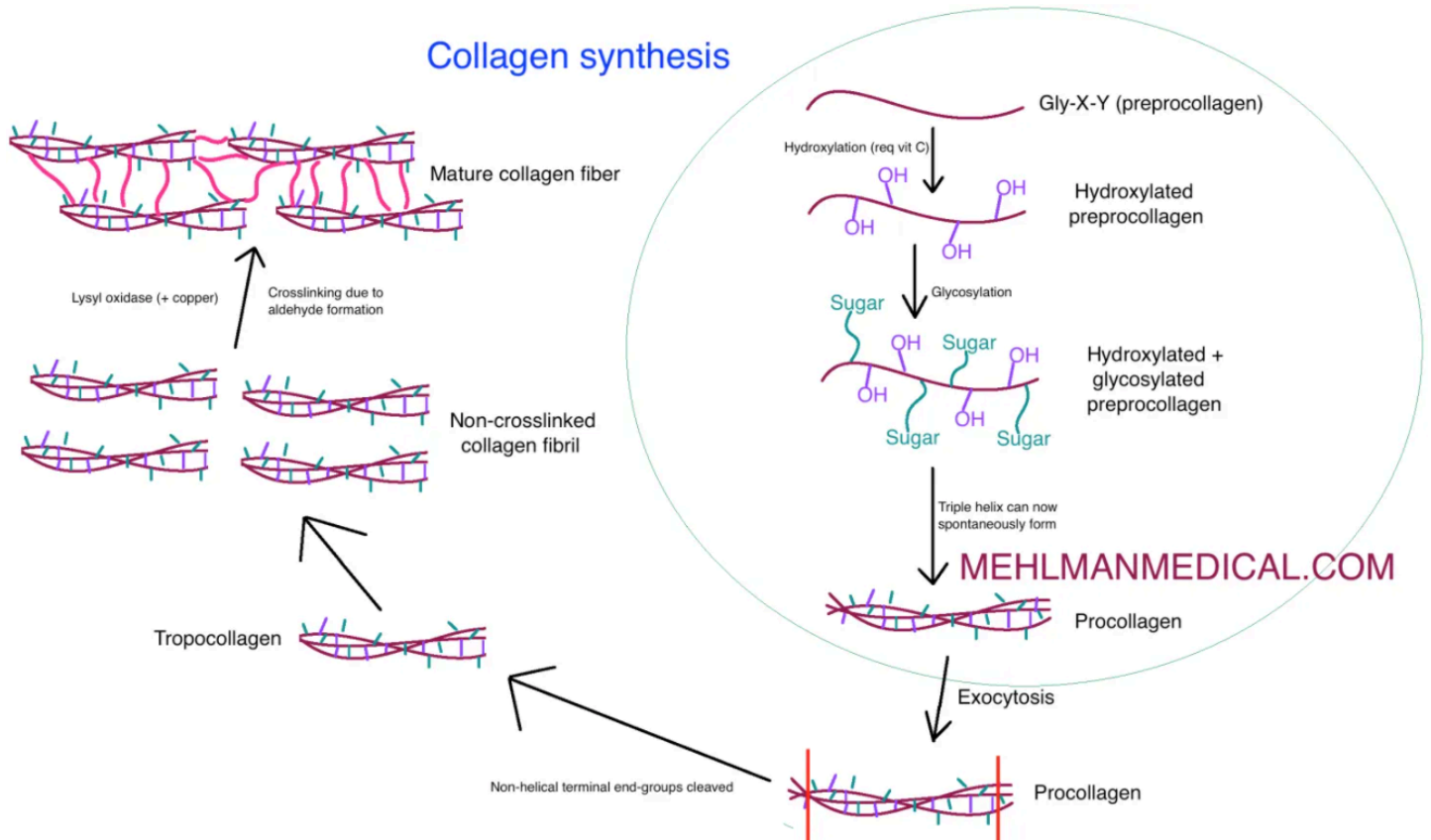
			- "Crumpled tissue paper" (lipid-laden) macrophages.
Tay-Sachs	Hexosaminidase A	GM2 ganglioside	- Cherry red spot on macula (blindness). - No hepatosplenomegaly. - Neurodegeneration.
Niemann-Pick	Sphingomyelinase	Sphingomyelin	- Cherry red spot on macula (blindness). - Yes, hepatosplenomegaly. - Niemann-Pick is a longer name than Tay-Sachs, and hepatosplenomegaly is a long word, so Niemann-Pick is the one with hepatosplenomegaly. - Neurodegeneration.
Metachromatic leukodystrophy	Arylsulfatase A	Cerebroside sulfate	- Progressive neurologic decline.
Krabbe	Galactocerebrosidase	Galactocerebroside, psychosine	- Aka Globoid cell leukodystrophy. - Globoid cells (giant, multinucleated cells). - Neurodegeneration.
Fabry (XR)	α -galactosidase A	Ceramide trihexoside	- Angiokeratomas (small, red/purple dots on the skin due to capillary dilation). - Heart and/or renal problems. - Can cause burning feet/hands ("Fabry crises").
Hurler	α -L-iduronidase	Glycosaminoglycans (GAGs) – i.e., Dermatan + heparan sulfates	- Gargoyle-like facies, stridor, aggressive behavior. - Yes, clouded corneas. - More severe form of Hunter.
Hunter (XR)	Iduronate sulfatase		- Gargoyle-like facies, stridor, aggressive behavior. - No clouded corneas. - "Hunters can see the X" → Therefore XR and no clouded corneas.

- Researcher is investigating the primary structure of a molecule that is defective in osteogenesis imperfecta; which amino acid is likely to be most abundant in this molecule? → answer = glycine; primary structure of collagen is Gly-X-Y, where glycine composes one-third of collagen; X and Y are usually proline or lysine, which are then frequently hydroxylated to become hydroxyproline and hydroxylysine. For some reason the USMLE is known to ask about collagen synthesis:

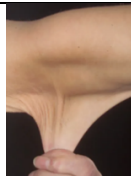
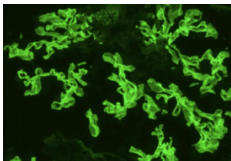
Collagen vs Elastin synthesis HY points	
Collagen	1) Collagen has a Glycine-X-Y (Gly-X-Y) primary protein structure, where the X and Y are often proline and hydroxyproline, as well as sometimes lysine and hydroxylysine. This Gly-X-Y sequence repeats. Therefore, collagen is 1/3 glycine. 2) Extra hydroxylations (-OH groups) are added. Vitamin C is needed as cofactor for hydroxylation. 3) Glycosylations (sugars) are added. 4) Then triple-helix formation occurs. 5) Terminal non-triple-helix end-groups are cleaved off. 6) Lysyl oxidase enables cross-linking of the collagen fibrils. 7) Can be degraded by matrix metalloproteinases. 8) Collagen demonstrates tensile strength, which prevents most stretching. If the question stem says anything about "tensile strength," you know they're talking about collagen. A HY point on USMLE is that if scar formation occurs, it is generally accepted that the tissue can regain ~80% of its original tensile strength.

Elastin	1) No Gly-X-Y structure. Contains desmosine and isodesmosine in addition to glycine, proline, and valine. Once again, there's no Gly-X-Y structure the way there is with collagen. 2) No extra hydroxylation and glycosylation steps. 3) No triple-helix formation. Instead forms an amorphous, cross-linked network. 4) Forms desmosine cross-linking. 5) Can be degraded by elastase. 6) High elasticity of elastin enables a return to original shape after stretching. This is enabled by the presence of alternating hydrophilic and hydrophobic domains, which allow elastin to act like a rubber band.
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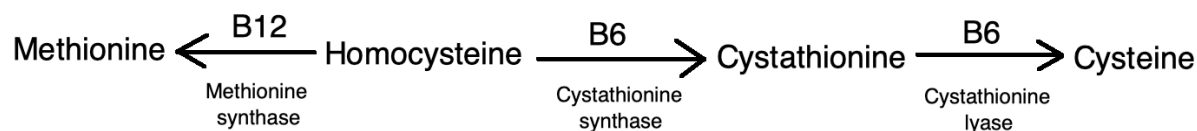
Collagen synthesis



Collagen disorders			
Collagen type	Where found	Disease(s)	HY points
I	Bone, Late wound healing (white in color), ↑ tensile strength compared to type III.	Osteogenesis imperfecta (OI)	- Fractures at different stages of healing. - Often mistaken for child abuse. - Blue sclerae (too easy; often omitted from Qs).  - Conductive hearing loss (malformation of ossicles). - If ruled out OI + child abuse, think osteopetrosis.
II	Cartilage, Intervertebral discs, vitreous humor.	Stickler syndrome	- Congenital hearing loss.
III	Blood vessels, Early wound healing (pink in color)	Ehlers-Danlos (vascular type)	- Hyperextensible skin/joints.

	↓ tensile strength compared to type I.		 - Easy bruising. - Aortic dissection/regurgitation. - Mitral valve prolapse. - Circle of Willis berry (saccular) aneurysms.
IV	Basement membrane (kidney, alveoli).	Alport, Goodpasture	- Alport (XR; <i>mutation in</i> type IV collagen); eye/ear problems in male with hematuria. - Goodpasture (<i>Abs against</i> type IV collagen); male 20s-40s with hemoptysis + hematuria; linear immunofluorescence pattern on biopsy. 

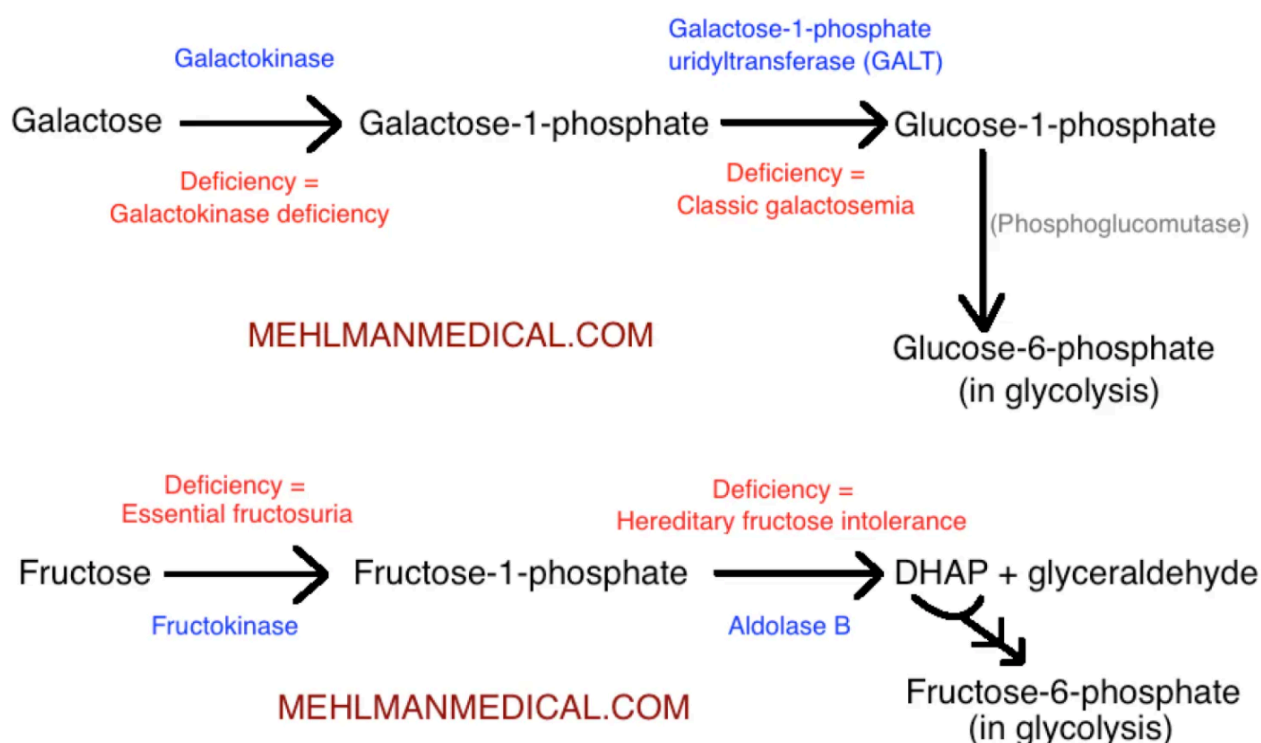
- 12F + tall and lanky + myopia on school visual acuity exam + thrombotic episode; Dx? → answer = homocystinuria (AR); usually due to deficiency of cystathionine synthase; homocystine = two homocysteines bound together; Q will be Marfanoid body habitus in school-age kid with thrombotic episode +/- a lens dislocation; B6 can be used as Tx in some patients.



- 13M + seizure + lens dislocation two years ago + serum cystathionine is decreased; what is most likely to be increased in this patient? → answer = methionine; shunting of homocysteine to methionine in this case.
- 31M + tall + flat feet + pectus excavatum + mid-systolic click; the abnormal molecule in this patient is best described as which of the following? → answer = “glycoprotein that forms a sheath around elastin” (fibrillin); Dx is Marfan syndrome; AD; *FBN1/2* genes on chromosome 15; fibrillin **is not** related to collagen (it stabilizes elastin); tall, lanky stature; mitral valve prolapse + aortic regurg (myxomatous degeneration) + aortic dissection (cystic medial necrosis) can be seen; scoliosis; arachnodactyly (long fingers); pes planus (flat feet); chest wall abnormalities (pectus excavatum/carinatum; abnormal hair distribution).

- **Sugar disorders:**

- Both galactose and fructose disorder vignettes may carry the descriptor of “reducing sugars” in the urine. Galactose disorders are detected by heel prick test at birth.
- Galactose disorders surface earlier than fructose disorders because of ingestion of milk (lactose = galactose + glucose).
- Fructose disorders don’t surface until infant commences fruit, juice, honey, and/or table sugar (sucrose = fructose + glucose).



Sugar disorders for USMLE	
- Both galactose and fructose disorder vignettes may carry the descriptor of “reducing sugars” in the urine. - Galactose disorders are detected by heel prick test at birth. - Galactose disorders surface earlier than fructose disorders because of ingestion of breast milk (lactose = galactose + glucose). - Fructose disorders don’t surface until infant commences fruit, juice, honey, and/or table sugar (sucrose = fructose + glucose).	
Condition	HY point(s)
Galactokinase deficiency	- Deficiency of galactokinase. LOL! - From birth (milk lactose → galactose + glucose); screened via heel-prick test. - Cataracts in neonate. Aldose reductase converts galactose → galactitol, which has ↑↑ osmotic pull in lens, leading to swelling and osmotic damage. - Milder than classic galactosemia.
Classic galactosemia	- Deficiency of galactose-1-phosphate uridylyltransferase. - From birth (milk lactose → galactose + glucose); screened via heel-prick test. - Cataracts in neonate. - Hepatomegaly, jaundice, hypoglycemia. - Associated with <i>E. coli</i> sepsis.

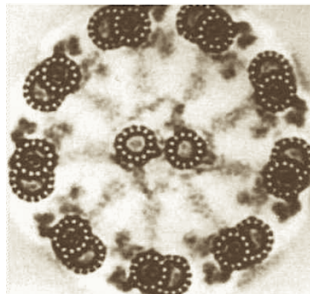
	- Trapped galactose-1-phosphate in liver sequesters / uses up phosphate. - Worse than galactokinase deficiency.
Essential fructosuria	- Deficiency of fructokinase. - Benign condition. - Mere presence of “reducing sugars” in the urine. - Negative glucose oxidase urine test (indicating reducing sugars are not glucose).
Hereditary fructose intolerance	- Deficiency of aldolase B. - Begins with dietary introduction of juice, fruit, honey, and table sugar. - Hepatomegaly, jaundice, hypoglycemia. - Trapped fructose-1-phosphate in liver sequesters / uses up phosphate. - Worse than essential fructosuria.

- Researcher is looking at a compound that is taken up by cells + no energy is required for uptake + the compound is not concentrated in the cell; Q asks, which mechanism best describes this mechanism of transport? → answer = carrier-mediated diffusion (facilitated diffusion/transport).

- Passive transport (no energy required):
 - Simple diffusion → Molecules move freely down concentration gradient (i.e., high to low) without any type of transporter. No energy is required. An example is oxygen and carbon dioxide moving into vs out of a cell, respectively.
 - Facilitated diffusion → Molecules move down concentration gradient (i.e., high to low) with the help of a transport protein. No energy is required because it follows the high-to-low concentration gradient. An example is GLUT transporters for sugars.
- Active transport (energy required):
 - Primary active transport → ATP is used directly to move molecules against their concentration gradients (i.e., low to high) using transport proteins. An example is ATPase pumps, such as for moving Na^+ and K^+ directly via the use of ATP.
 - Secondary active transport → Energy from an ion gradient is used to transport molecules. In this case, ATP is used to indirectly to transport molecules by first creating the ion gradient, and then energy from that gradient transports the molecules. An example is ENaC on apical membrane of the renal cortical collecting duct. Sodium moves down its concentration gradient via ENaC from the urine into the cell, but this is only possible because a favorable high-to-low gradient is first

established via a Na^+/K^+ ATPase pump on the basolateral membrane, as stimulated by aldosterone.

- 35F + recurrent bronchitis and sinusitis throughout life + Hx of ectopic pregnancy + cardiac exam shows point of maximal impulse at the 4th intercostal space right midclavicular line + biopsy of nasal polyp taken; Q asks, which structure is most likely to be absent on electron microscopy of the biopsy specimen? → answer = dynein arms; Dx is primary ciliary dyskinesia (Kartagener syndrome); dynein arms are required for function of cilia, which normally line the pseudostratified columnar epithelium of the respiratory tree → decreased function of cilia → increased respiratory tract infections; Kartagener is associated with situs inversus / dextrocardia; ↑ ectopic pregnancies in females (cilia normally line Fallopian tubes) and ↓ sperm motility in males (infertility); a cilium comprises a 9x2 circumferential pattern of microtubules on 2D-cross-section (i.e., a cilium is much larger than microtubules); should be noted that whilst sperm demonstrate ↓ motility in Kartagener, there are *absent* sperm in cystic fibrosis due to CBAVD (congenital bilateral absence of vas deferens) – i.e., recurrent sinopulmonary infections + absent sperm = CF; whereas recurrent sinopulmonary infections + immotile sperm = Kartagener. Cross-section of cilium showing 9x2 microtubules circumferentially:



- 6M + 3rd percentile for weight and height + failed to pass meconium at birth; which of the following molecular combinations best describes the defective channel inherent to this patient' disease? → answer = cAMP and ATP; diagnosis is cystic fibrosis (AR; chromosome 7); CFTR is a cAMP-mediated, ATP-gated chloride channel; the defective channel is retained within the RER of the cytosol (i.e., it is **not** found on the cell membrane); sweat chloride test demonstrating >60 mEq/L is *more accurate* than genotyping due to ↑↑ allelic heterogeneity (i.e., many diseased alleles can produce the CF phenotype; even detailed genetic screening panels only detect upward of 90-96% of the most common mutations); ΔF508 (deletion of phenylalanine at position 508) is most common CF mutation;

patients have negative nasal transepithelial potential difference (one type of diagnostic test); *Staph aureus* exceeds *Pseudomonas* for most common cause of CF pneumonia in first decade of life; after first decade, *Pseudomonas* exceeds *Staph*; phenotypically normal sibling of patient with CF has 2/3 chance of being a carrier (i.e., we can eliminate the aa on the Punnett square because the unaffected sibling clearly doesn't have the disease, so 2Aa / 2Aa + 1AA); dornase-alfa (correct, not alpha) is a nucleotidase that can help clear mucous from respiratory tree; N-acetylcysteine has -SH groups that can help breakdown mucous; guaifenesin softens mucous; ivacaftor is CFTR modulator that can help re-fold protein into functional form.

- 24F + autosomal recessive condition + low serum vitamin D + Hx of two episodes of *Pseudomonas* pneumonia; what is most likely to occur on a cellular level in this patient's pancreatic ducts? → answer = "upregulation of ENaC"; Dx is cystic fibrosis; CFTR channel is defective → Cl⁻ cannot be secreted out of pancreatic ducts into the lumen → more Cl⁻ is retained in the cell → more Na⁺ moves from lumen into the cell to balance charge via ENaC → water follows Na⁺ → secretions within pancreatic duct lumen inspissate (i.e., desiccate within a lumen) → exocrine pancreas malsecretion → impaired absorption of fat-soluble vitamins and macronutrients → failure to thrive.
 - Should be noted that chloride moves the opposite direction in sweat glands – i.e., rather than CFTR functioning to secrete Cl⁻, it reabsorbs it, hence our positive sweat chloride test.
- 17F + taking OCPs + taking isotretinoin for acne past two years + has headache + is vegan; which vitamin is most likely responsible for her headache → answer = vitamin A → isotretinoin (high-dose vitamin A) and OCPs both can cause pseudotumor cerebri (increased intracranial pressure); veganism can cause B12 deficiency, not excess, nor does that relate to headaches.
- Investigator is looking at enzyme that is a giant complex; which of the following is most likely? → answer = pyruvate dehydrogenase (converts pyruvate → acetyl-CoA of TCA cycle); described as giant complex of vitamins B1, B2, B3, B5, and lipoic acid.

HY vitamins and minerals for USMLE	
- The fat-soluble vitamins are A, D, E, K (All Dogs Eat Kittens). - All vitamin D-related endocrine relations are pass-level for the Steps, so know these really well. -	
Nutrient	HY points
Vitamin A	- Different compounds – i.e., retinol, retinal, retinoic acid. - Acts as transcription factor by binding to retinoic acid receptor in the nucleus, which then upregulates gene transcription.

	- Required for night vision, skin turnover, and immune function. - Topical tretinoin is first-line for acne. It shuts off sebum production and increases epithelial cell turnover. - Oral isotretinoin (high concentration vitamin A) is last-line because of its adverse effects but is most effective for acne. - Both of these treatments cause photosensitivity and desquamation (peeling of skin). - High-dose vitamin A is teratogenic, so must do beta-hCG before starting isotretinoin. Some Qs will also advise commencing birth control before use, or will require educating the patient about barrier contraception, etc. - If USMLE asks most likely teratogenicity in Q about vitamin A, choose cleft lip/palate. Topical tretinoin is not teratogenic. - Vitamin A excess causes cerebral edema with pseudotumor cerebri (i.e., high intracranial pressure, papilledema, confusion), hepatotoxicity (increased liver enzymes), and dyslipidemia. - Isotretinoin and ursine (bear) are the two most common causes of vitamin A toxicity. - Deficiency causes nyctalopia, squamous metaplasia, and immunodeficiency. - Deficiency is rare because Vitamin A is fat-soluble and also stored in high amounts in the liver in Ito cells. - Deficiency caused by malabsorptive disorders - i.e., CF, chronic pancreatitis, Crohn disease. - Vitamin A (all-trans retinoic acid) can be used as treatment for some types of acute myelogenous leukemia (AML).
Vitamin B1 (thiamine)	- Used for dehydrogenase reactions. The four B1-dependent enzymes for USMLE are: 1) Pyruvate dehydrogenase (pyruvate → acetyl-CoA). 2) Alpha-ketoglutarate dehydrogenase (alpha-KG → succinyl-CoA in TCA cycle). 3) Branched-chain ketoacid dehydrogenase (deficient in maple syrup urine disease, which causes inability to metabolize leucine, isoleucine, valine). 4) Transketolase (an RBC enzyme that requires thiamine and can be used to test for thiamine deficiency, since transketolase activity goes ↓ in B1 deficiency). - Deficiency is classic in alcoholics and causes Wernicke-Korsakoff syndrome and dry and wet beri beri. - Wernicke-Korsakoff syndrome (combo of Wernicke encephalopathy + Korsakoff psychosis) occurs due to mammillary body dysfunction in B1 deficiency. - Acronym for Wernicke encephalopathy → A COW → Ataxia/Anterograde amnesia, Confusion, Ophthalmoplegia, Wernicke. - B1 is given with glucose in emergency setting to unconscious patients or to known alcoholics. If the Q asks what giving B1 to an alcoholic patient is most likely to prevent, the answer is any of the ACOW findings, e.g., anterograde amnesia. - Korsakoff psychosis is retrograde amnesia that presents as confabulations, which means the patient makes up details about their past / what they did that day because they don't remember. If the Q doesn't mention confabulations in a patient with ACOW, the patient only has Wernicke encephalopathy, not Wernicke-Korsakoff syndrome. - Dry beri beri = neuropathy; can present with (+) Romberg sign. - Wet beri beri = dilated cardiomyopathy. - Deficiency can also be seen, oddly enough, in gastric bypass patients who have confusion and neuropathy. This shows up on two new NBME Qs, where B12 deficiency is wrong. Presumably this is because confusion is part of Wernicke for thiamine deficiency, whereas B12 deficiency doesn't classically cause confusion. So the neuropathy refers to dry beri beri due to B1 deficiency rather than subacute combined degeneration, which is the name of the neuropathy in B12 deficiency.
Vitamin B2 (riboflavin)	- Used as a cofactor in FAD/FMN reactions. - Required for succinate → fumarate in TCA cycle, where FAD → FADH₂. - B2 and B6 used as cofactors in niacin (B3) synthesis.
Vitamin B3 (niacin)	- Used for NAD⁺/NADH reactions; Q might give you patient with B3 deficiency and then have you select "adenine" as the answer. B3 is part of a cofactor that contains adenine. - HY cardiovascular effects: ↑↑ HDL; ↓ VLDL export by the liver.

	- Tryptophan is a precursor to niacin synthesis. Hartnup disease ($\downarrow$ ability to reabsorb tryptophan in PCT of kidney) leads to niacin deficiency. - Tryptophan is also precursor to serotonin. Carcinoid syndrome consumes tryptophan, since serotonin is produced, leading to niacin deficiency in some cases. - Deficiency of B3 causes pellagra (dementia, dermatitis, diarrhea). Can present with delirium (biggest risk factor for delirium is underlying dementia). The dermatitis will be either Casal necklace, hyperpigmentation of forearms, or blistering. - B3 excess causes hyperinsulinemia (insulin resistance and diabetes), hyperuricemia (gout), and flushing (due to prostaglandin production, not histamine). Aspirin/NSAIDs can be used to prevent/mitigate flushing, so it makes sense that prostaglandin, not histamine, is the cause. - Q will tell you guy has big cardiovascular history and podagra (gout of big toe; don't confuse with pellagra) and then ask which agent is contraindicated $\rightarrow$ answer = niacin (since can cause gout).
Vitamin B5	- Aka pantothenic acid. - Precursor to coenzyme-A (transferred to pyruvate to make acetyl-CoA). - One of the cofactors for pyruvate dehydrogenase, which requires B1, B2, B3, B5, and lipoic acid as cofactors.
Vitamin B6	- Aka pyridoxine. - First step in heme synthesis: - Succinyl-CoA + glycine, via B6 and δ-ALA synthase $\rightarrow$ δ-ALA. HY for USMLE. - Used for decarboxylase reactions, e.g., L-Dopa $\rightarrow$ dopamine via L-Dopa decarboxylase. - Transaminase reactions in the liver, i.e., the ALT and AST enzymes. - Used to synthesize cysteine from homocysteine (B6 can be given to some patients with homocystinuria). - Homocystinuria is the answer when they give you a Marfanoid body habitus (means looks like Marfan syndrome but isn't) in a patient with visual changes and sometimes history of thrombosis. In other words: - Homocystinuria = tall, lanky patient with changes in vision and thromboses. - Deficiency caused by isoniazid (patients with TB) $\rightarrow$ presents as peripheral neuropathy +/- seizures. Vitamin B6 must be given to patients on isoniazid.
Vitamin B7	- Aka biotin. - Used for carboxylase reactions. The important one for USMLE is: - Pyruvate $\rightarrow$ oxaloacetate (OAA) via pyruvate carboxylase. - This is a gluconeogenic enzyme, where pyruvate $\rightarrow$ OAA, which then goes back to PEP (the second to last substrate in glycolysis), and then PEP goes back up to glucose. - In other words, instead of pyruvate $\rightarrow$ acetyl-CoA for use in the TCA cycle via pyruvate dehydrogenase in the fed state (i.e., when insulin is high), pyruvate carboxylase will be active in the fasting state (i.e., when insulin is low). - Pyruvate carboxylase is used in biochemistry arrow Qs on USMLE, where they'll give you a scenario (i.e., fed or fasting, or insulin high vs low, etc.), and then you have to select an up or down arrow for pyruvate carboxylase activity. - Deficiency caused by eating raw egg whites (contain avidin, which binds B7).
Vitamin B9	- Aka folate/folic acid. Normally found in dark green, leafy vegetables. - Used in an array of complex reactions with B12. - Required for synthesizing thymine from uracil. An enzyme called thymidylate synthase requires B9. 5-fluorouracil is a thymidylate synthase inhibitor. - Most common vitamin deficiency, which results in some HY findings: - Megaloblastic anemia and hypersegmented neutrophils. - Neural tube defects (critical during first 3-4 weeks' gestation). - HY agents that cause B9 deficiency: antiepileptics (i.e., valproic acid, phenytoin, carbamazepine) and methotrexate. - The Q will give you an image of a hypersegmented neutrophil in someone on phenytoin, and the answer is B9 deficiency, for instance, not B12.

	- Deficiency causes: $\uparrow$ homocysteine and no change methylmalonic acid. In B12 deficiency, both homocysteine and methylmalonic acid are $\uparrow$.
Vitamin B12	- Aka cyanocobalamin. Found in animal products, fish, and shellfish. - Used in an array of complex reactions with B9. - Required by methionine synthase to synthesize methionine from homocysteine. - Required by methylmalonyl-CoA mutase (methylmalonyl-CoA $\rightarrow$ succinyl-CoA). - Deficiency causes megaloblastic anemia and hypersegmented neutrophils (same as with B9), however unique to B12 deficiency is a type of neuropathy called subacute combined degeneration, which is demyelination of corticospinal tracts, dorsal columns, and spinocerebellar tracts (<i>spinothalamic not involved</i>) - Subacute combined degeneration is caused by the buildup of methylmalonyl-CoA, which is toxic to myelin. - Deficiency causes: $\uparrow$ homocysteine and $\uparrow$ methylmalonyl-CoA (or methylmalonic acid); in B9 deficiency, only homocysteine is $\uparrow$. - Deficiency caused by various HY etiologies: - Loss of parietal cells (i.e., pernicious anemia, gastric bypass surgery, chronic gastritis). Parietal cells make intrinsic factor, which is required for absorption of B12 at the terminal ileum. - Terminal ileum resection or absorptive disorders affecting the terminal ileum, such as Crohn disease. - Veganism or strict vegetarianism. - <i>Diphyllobothrium latum</i> (fish tapeworm) infection. - Schilling test can be used to ascertain etiology of deficiency. - Low B12 can be theoretical cause of cognitive decline in elderly. In an elderly patient with "dementia," you must screen for pseudodementia (depression that presents like dementia), B12 deficiency, hypothyroidism (check TSH), neurosyphilis (if sexually active, even in elderly yes), and neurolyme (if in endemic area).
Vitamin C	- Aka ascorbic acid. - Used in hydroxylase reactions. - This is relevant to collagen synthesis, especially hydroxylation of proline residues after the Gly-X-Y primary structure is formed. - Norepinephrine synthesis (dopamine $\rightarrow$ NE, via dopamine β-hydroxylase). - Plays a role in immune function. - Required for iron absorption in duodenum (vitamin C ferrireductase). - Deficiency (scurvy): bleeding gums, perifollicular hemorrhages, ill-appearing patient. - Excess can cause calcium oxalate stones. More obscure cause, but possible.
Vitamin D	- Required for calcium homeostasis and bone calcification. - Synthesis starts in stratum basale of epidermis as 7-dehydrocholesterol. USMLE will specifically ask you which layer, so know stratum basale. - 7-dehydrocholesterol, via UVB radiation $\rightarrow$ cholecalciferol. If the USMLE Q tells you someone doesn't get sunlight, the answer is deficiency of cholecalciferol (<i>those who don't get sunlight still make 7-dehydrocholesterol</i>). - Cholecalciferol goes to the liver, where it's hydroxylated to make 25-OH-D3 (calcidiol). If the Q gives you an alcoholic, then 25-OH-D3 will be $\downarrow$, where the answer is "decreased hepatic hydroxylation." - 25-OH-D3 then goes to the PCT of the kidney, where PTH upregulates 1α-hydroxylase to convert 25-OH-D3 into $1,25-(OH)_2-D3$ (calcitriol). - $1,25-(OH)_2-D3$ then goes to the small bowel where it causes Ca and PO_4 absorption. - $1,25-(OH)_2-D3$ also goes to bone where it converts unmineralized osteoid into mineralized hydroxyapatite. - Vitamin D deficiency causes rickets (children) and osteomalacia (adults). - Deficiency caused by malabsorptive disorders – e.g., CF, chronic pancreatitis, Crohn. - For deficiency, choose $\downarrow$ Ca, $\downarrow$ PO_4, $\uparrow$ PTH (goes up to compensate for $\downarrow$ Ca). - Mechanism for $\downarrow$ Ca in renal failure is a combination of 1) not being able to reabsorb Ca in the late-DCT under the action of PTH, and 2) $\downarrow$ PCT 1α-hydroxylation means $\downarrow$ small bowel absorption of Ca.

	- Hypervitaminosis D (vitamin D excess) can occur due to granulomatous diseases. On USMLE this is always sarcoidosis → epithelioid macrophages within non-caseating granulomas produce 1α-hydroxylase without the need for PTH → calcitriol increases → ↑ small bowel absorption of Ca → PTH is suppressed. - For high vitamin D (sarcoidosis), select ↑ Ca, ↑ PO₄, ↓ PTH. - Williams syndrome → AD, chromosome 7, elfin-like facies, supravalvular aortic stenosis, hypercalcemia secondary to increased vitamin D sensitivity.
Vitamin E	- Aka tocopherol (i.e., tocopherol acetate). - Free-radical scavenger. USMLE can write this as “decreases membrane peroxidation.” - Deficiency is rare but associated with neuropathy similar to the subacute combined degeneration seen in B12 deficiency. This is particularly HY in cystic fibrosis Qs, as per my observation across NBMEs. - Excess can antagonize vitamin K, leading to increased bleeding propensity.
Vitamin K	- Cofactor for the enzyme gamma-glutamyl-carboxylase → activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. - Do not confuse gamma-glutamyl-carboxylase with vitamin K epoxide reductase. The latter is inhibited by warfarin, but vitamin K is merely a cofactor for the former. Vitamin K epoxide reductase recycles vitamin K to its active form. - Deficiency will present in 3 ways on USMLE: 1) Neonate with bleeding from umbilical stump (sterile bowl in neonate has not yet synthesized vitamin K; neonate needs Vit K injection). 2) Older patient on broad-spectrum Abs for 6+ weeks, leading to obliteration of gut flora. This can be in response to the Q asking why the patient needs ↓ warfarin dose. Since warfarin inhibits vitamin K function, if we have ↓ vitamin K, we also simply require ↓ warfarin to achieve the same INR. 3) Patient who was exposed to rat poison, which apparently has a warfarin-like effect. - Vitamin K is the slow reversal agent for warfarin. If patient is actively bleeding or requires surgery, give fresh frozen plasma instead to quickly reverse warfarin.
Copper	- Essential cofactor for lysyl oxidase, which catalyzes crosslinking collagen. - Excess copper seen with Wilson disease → AR, chromosome 13; inability to secrete copper into bile; leads to ↑ copper in urine + ↓ serum ceruloplasmin; causes cirrhosis, Parkinsonism (Cu deposition in basal ganglia, especially the putamen), Kayser-Fleischer rings (deposition in Descemet membrane of cornea); Tx is penicillamine (copper chelator). - Deficiency seen in Menkes disease → XR; inability to absorb Cu in duodenum; causes “wiry” or “springy” hair.
Iron	- Main use is in hemoglobin. If you didn’t know this basic point, you’re f-u-c-k-e-d-d-d. - Excess seen in hereditary hemochromatosis → AR, chromosome 6; leads to increased duodenal absorption of iron (i.e., cannot shut off absorption); causes Bronze diabetes → hyperpigmentation due to hemosiderin, diabetes due to Fe deposition in tail of pancreas, and various other findings such as arthritis (pseudogout), infertility, and cardiomyopathy; Tx is serial phlebotomy; chelators are the wrong answer in hereditary hemochromatosis; deferoxamine or deferasirox are chelators given in transfusional siderosis (secondary hemochromatosis due to chronic blood transfusions).
Zinc	- Used in zinc finger domains, which are part of some transcription factors (i.e., zinc plays a role in DNA binding); zinc found in high concentrations in semen. - Deficiency causes anosmia (↓ smell), hypogeusia (↓ taste), alopecia, rash, and diarrhea. Harder NBME Q doesn’t mention ↓ smell and taste, and only mentions alopecia, rash, and diarrhea. So don’t rely on the ↓ smell and taste as mandatory. - Acrodermatitis enteropathica → obscure AR disorder where there’s inability of jejunum to absorb zinc; presents with dermatitis.

- 42M + alcoholic + high MCV; Q asks, which combination of serum methylmalonic acid and homocysteine is most likely in this patient? → answer = normal methylmalonic acid (or methylmalonyl-CoA), ↑ homocysteine; diagnosis is folate (B9) deficiency. Learning point is:
 - B9 deficiency: normal methylmalonic acid (or methylmalonyl-CoA), ↑ homocysteine.
 - B12 deficiency: ↑ methylmalonic acid (or methylmalonyl-CoA), ↑ homocysteine.
 - This is because without B9 and B12, there is less homocysteine → methionine.
 - However B12 alone is needed to convert methylmalonyl-CoA into succinyl-CoA.
- Experiment performed showing HPV16 E6 protein causes p53 protein degradation; Q asks, which cellular enzyme will target p53 as a result of this viral protein? → answer = ubiquitin ligase; ubiquitination will target proteins for degradation by the proteasome.
- 31F + Pap smear shows atypical squamous cells + molecular diagnostic studies show viral E6 protein; Q asks, this protein promotes cell growth and malignancy via what mechanism? → answer = “degradation of p53”; HPV16 E6 protein targets p53; HPV18 E7 protein targets RB; both are tumor suppressor proteins; you can remember this because 16 comes before 18, as 6 comes before 7, as P comes before R.
- 62M + back pain + restrictive cardiomyopathy + renal failure + biopsy of kidney shows a birefringent pattern with Congo red stain; Q asks, what is the structure of the material that is stained? → answer = β-pleated sheet; diagnosis is multiple myeloma causing cardiac and renal amyloidosis; amyloid stains apple green birefringent with Congo red stain because of its β-pleated sheet structure.
- Neonate + several fractures occurred during birth + shortened extremities + misshapen long bones + poor skull mineralization; Q asks, this girl has a defect in which of the following? → answer = collagen; diagnosis is osteogenesis imperfecta (collagen I).
- Experiment with fish embryos; two pronuclei used to make embryo; results of study show that if both pronuclei are of maternal or paternal origin, the embryo fails to develop properly; why? → answer = imprinting; normally two alleles are inherited for each gene an organism has (one from each parent), but various alleles will sometimes be silenced from one parent and expressed in the other; if the allele coming from mom is silenced, that’s called maternal imprinting; if the allele coming from dad is silenced, that’s called paternal imprinting.

- Prader-Willi syndrome (PWS; many involved genes; mental retardation + hyperphagia) is classic maternal imprinting → gene coming from mom is normally silenced; dad's gene is supposed to be expressed but is deleted or mutated ("**Willi hates his dad**" because dad's gene is deleted); even if mom's gene is healthy, since it is normally silenced, it will not be expressed.
- Angelman syndrome (AS; *UBE3A* gene on chromosome 15; "happy puppet") is classic paternal imprinting → gene coming from dad is normally silenced; mom's gene is supposed to be expressed but is deleted or mutated ("**Mom is not an angel**" because mom's gene is deleted); even if dad's gene is healthy, since it is normally silenced, it will not be expressed.
- PWS and AS can also be caused by uniparental disomy, where both alleles are erroneously inherited from the same imprinted parent.
 - For PWS (maternal uniparental disomy), since the gene is normally maternally imprinted, if both alleles are inherited from mom, both will be silenced; disease occurs because there's no paternal allele available to be expressed.
 - For AS (paternal uniparental disomy), since the gene is normally paternally imprinted, if both alleles are inherited from dad, both will be silenced; disease occurs because there's no maternal allele available to be expressed.
- 23M + develops pancreatic cancer + mom had colon cancer + uncle had liver cancer; investigative studies show mutation in *TP53* gene; Q asks, which process accounts for difference in presentation among family members; answer = pleiotropy; diagnosis is Li-Fraumeni syndrome (*TP53* gene mutation results in p53 protein abnormality).
 - **Pleiotropy** refers to unrelated phenotypes resulting from a mutation in a single gene. A HY USMLE example is unrelated types of cancers in family members with Li-Fraumeni syndrome. Phenylketonuria (PKU) is another HY example (i.e., mutation can result in partial albinism and mental retardation, which are unrelated phenotypic traits). Pleiotropy need not refer to different traits phenotypically expressed in the same individual.
 - **Variable expressivity**, in contrast, does not specifically refer to the aspect of multiple, disparate phenotypes caused by a single gene, but instead refers to differences in *disease*

severity among individuals with a condition that demonstrates a predictable spectrum of findings. For instance, NF1 is a HY example of a condition with both variable expressivity *and* pleiotropy. One family member might have more severe disease with café au lait spots, pheochromocytoma, neurofibromas, and Lisch nodules (iris hamartomas), whereas another family member has milder disease with only axillary/groin freckling; the difference in *disease severity* between family members along the predictable spectrum of possible features of NF1 refers to variable expressivity. The pleiotropic aspect refers to mutations in NF1 merely causing different phenotypes, such as pheochromocytoma versus café au lait spots. Conditions that demonstrate variable expressivity must demonstrate complete (100%) penetrance as part of the definition.

- **Penetrance** refers to the percentage of those with a diseased genotype who demonstrate the diseased phenotype to any degree. In other words, if 100% of those with a diseased genotype express the disease phenotypically in some form, then the disease is said to be completely penetrant (NF1 is an example). If the diseased genotype does not result in a disease phenotype in all individuals, then the disease is said to be incompletely penetrant (BRCA1/2 causing breast cancer that “skips a generation” is an example).
- 40M + pancreatic cancer due to Li-Fraumeni syndrome; Q asks, what mechanism is responsible for development of cancer in this patient? → answer = “loss of heterozygosity”; patient was born with one diseased allele in every cell (i.e., one hit); after spontaneous mutation in second allele, disease occurred. The p53 protein is a tumor suppressor that halts the cell cycle in the setting of DNA damage; sometimes the Q will simply have “failure of DNA repair” as the answer for Li-Fraumeni.
- 44M + cognitive decline over the past six months + abnormal movement of limbs + father had similar presentation in his 50s; Q asks, what mechanism best explains this patient’s presentation? → answer = anticipation; Dx is Huntington disease (AD, chromosome 4), which is caused by 40+ trinucleotide repeats (TNR) of CAG codon; anticipation is a HY term associated with TNR disorders that refers to the disease presenting increasingly more severe, and earlier, with each successive generation. This is due to expansion of the number of TNR with each generation.

Trinucleotide repeat (TNR) disorders	
Condition	HY points
Huntington	- AD; chromosome 4; >40 CAG repeats = disease. - Encodes huntingtin protein (yes, different spelling). - Neurodegeneration with cognitive decline and choreoathetosis beginning in 40s or 50s. - Anticipation is expansion of TNR length with each generation, resulting in earlier and more severe onset of disease.
Friedreich ataxia	- AR; chromosome 9; GAA expansion. - Encodes frataxin protein. - Ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, early-onset type II diabetes.
Fragile X	- X-linked; CGG expansion on <i>FMR1</i> gene. - CGG → Fra^Gile X. - The TNR expansion causes gene hypermethylation. This mechanism is asked on NBME. - Low IQ (most common hereditary cause after Down syndrome); macroorchidism, long, narrow jaw/face with everted ears.
Myotonic dystrophy	- AD; chromosome 19; CTG expansion. - CTG → Myo^Tonic dystrophy. - Encodes myotonic dystrophy protein kinase. - ↓ ability to relax muscles (i.e., letting go of doorknob, golf club, handshake).

Serum biochemistry HY points:

- Calcium:

- Normal range: 8.4-10.2 mg/dL.
- Low calcium causes “up” presentations - i.e., tetany and hyperreflexia. It can cause long QT on ECG.
- High calcium causes “down” presentations of muscle flaccidity and hyporeflexia. It also is associated with delirium/confusion (hypercalcemic crisis) and short QT on ECG.
- **High calcium HY causes:**
 - Sarcoidosis --> due to high vitamin D.
 - Primary hyperparathyroidism --> adenoma, hyperplasia, or MEN 1/2A.
 - Metastatic malignancy --> mets cause cytokine effects at bone, leading to lytic lesions and calcium leaching out of bone.
 - Multiple myeloma --> lytic lesions.
 - Thiazides --> cause increased calcium reabsorption in DCT.
- **Low calcium HY causes:**
 - Rickets/osteomalacia --> deficiency of vitamin D --> decreased small bowel Ca absorption.

- Secondary hyperparathyroidism (renal failure) --> decreased synthesis of 1,25-D3 --> decreased small bowel absorption.
- Post-thyroidectomy --> causes concomitant loss of parathyroids --> less calcium resorbed from bone + less 1,25-D3 synthesized in the kidney.
- DiGeorge syndrome --> agenesis of 3rd and 4th pharyngeal pouches --> agenesis of parathyroid glands.
- Loop diuretics --> cause decreased paracellular Ca reabsorption.
- Hypomagnesemia --> causes hypocalcemia and hypokalemia non-responsive to supplementation; usually seen in alcoholics.

- **Sodium:**

- Normal range: 135-145 mEq/L.
- Sodium derangement (high and low) causes CNS dysfunction --> confusion, stupor, or coma.
- Correcting high sodium too quickly with hypotonic saline can cause cerebral edema.
Correcting low sodium too quickly with hypertonic saline can cause central pontine myelinolysis (osmotic demyelination).
- **High sodium HY causes:**
 - Primary hyperaldosteronism --> aldosterone pulls sodium out of the urine at the cortical collecting duct.
 - Renal artery stenosis and fibromuscular dysplasia --> RAAS is increased --> higher aldosterone.
 - Dehydration --> hemoconcentration of sodium. RAAS would also be increased.
 - Diabetes insipidus --> cannot reabsorb free water at medullary collecting duct --> serum concentrates as a result.
- **Low sodium HY causes:**
 - Addison disease and Waterhouse-Friderichsen syndrome --> aldosterone is low.
 - Psychogenic polydipsia --> dilutional hyponatremia from excessive water consumption.

- DKA --> dilutional hyponatremia from high glucose in the blood --> water stays with glucose.
- SIADH --> reabsorbing too much free water at the medullary collecting duct --> serum dilutes.

- **Potassium:**

- Normal range: 3.5-5.0 mEq/L.
- Potassium derangement (both high and low) causes arrhythmia and muscle weakness.
- Hyperkalemia causes peaked T waves on ECG. For hyperkalemia with ECG changes, give IV calcium to stabilize the myocardium.
- Hypokalemia can cause U waves or flattened T waves on ECG.
- **High potassium HY causes:**
 - Addison disease and Waterhouse-Friderichsen syndrome --> aldosterone is low, so less K⁺ is secreted.
 - Renal failure --> K⁺ can't be filtered sufficiently at the glomerulus + cannot be secreted distally sufficiently under the action of aldosterone.
 - DKA --> insulin isn't present to push potassium into cells + "cellular shift" occurs, where cells take on more H⁺ from ketoacids, so they kick out K⁺ to balance charge. It should be noted, however, that low *total body* potassium occurs in DKA, despite high serum potassium (hyperkalemia). In other words, there is low intracellular K⁺, but high serum K⁺. Overall, total body K⁺ is low.
 - Polypharmacy with agents such as potassium-sparing diuretics (spironolactone, eplerenone, amiloride, triamterine) and digoxin.
 - Congenital adrenal hyperplasia caused by 21-hydroxylase deficiency.
- **Low potassium HY causes:**
 - Primary hyperaldosteronism (Conn syndrome and hyperplasia).
 - Renal artery stenosis + fibromuscular dysplasia --> high aldosterone.
 - Cushing disease (chronically high glucocorticoid levels can cause hypo-K due to distal renal secretion similar to mineralocorticoids).

- Vomiting + diarrhea --> cause volume depletion --> RAAS increases --> aldosterone kicks out K⁺. There is also direct loss from GI contents.
- Eating disorders (anorexia, bulimia), even if there isn't purging. Most common cause of death in these disorders is low potassium leading to arrhythmia.
- Low *total body* potassium occurs in DKA, despite high serum potassium (hyperkalemia).
- Loop diuretics --> direct loss due to Na/K/2Cl ATPase blockade + indirect loss from volume depletion that leads to increased RAAS.
- Thiazides --> cause volume depletion --> increased RAAS.
- Hypomagnesemia --> hypomagnesemia can cause hypokalemia and hypocalcemia non-responsive to supplementation; usually seen in alcoholics due to dietary deficiency. Mg is required for PTH release from the parathyroid glands. Mg is also required to prevent renal potassium wasting.

- **Phosphate:**

- Normal range: 3-4.5 mg/dL.
- **High phosphorus HY causes:**
 - Hypoparathyroidism --> kidney retains phosphate at the PCT, since PTH normally causes PO₄ loss.
 - Secondary hyperparathyroidism (renal failure) --> kidney cannot filter/excrete enough PO₄.
 - Tumor lysis syndrome --> cells release lots of PO₄ into blood + acute kidney failure can occur, resulting in high PO₄ by extension.
 - Sarcoidosis --> high 1,25-D₃ causes increased small bowel absorption of phosphate (and calcium).
- **Low phosphorus HY causes:**
 - Primary hyperparathyroidism --> increased phosphate loss at the PCT of the kidney.
 - Rickets/osteomalacia --> low vitamin D means less absorption of PO₄ from small bowel. It should be noted, however, that if the osteomalacia is specifically due to

renal failure, then phosphate is still up, since the effect of the renal failure "wins" over the low vitamin D effect.

- Refeeding syndrome --> phosphate is consumed by the abrupt increase in glycolytic intermediates requiring phosphate.

- **Magnesium:**

- Normal range: 1.7-2.2 mg/dL.
- Magnesium derangement causes effects similar to calcium – i.e., low Mg presents with an “up” state of hyperreflexia and increased muscle tone; high Mg presents with a “down” state of hyporeflexia and muscle flaccidity.
- Low Mg HY cause is alcoholism (decreased dietary intake) --> leads to hypocalcemia and hypokalemia non-response to supplementation.
- High Mg doesn't really exist on USMLE.

- **HY Acid-base concepts for USMLE:**

- **Normal serum pH: 7.35-7.45.**
- **Normal serum bicarb (HCO_3^-): 22-28 mEq/L.**
- **Normal arterial pCO_2 : 33-44 mmHg.**

- If the Q gives you a respiratory disturbance of any kind, look at pCO_2 first in the labs.
- If the Q doesn't give you a respiratory disturbance, then the cause of the acid/base disturbance is metabolic, so look at HCO_3 first.
- CO_2 is acidic, so if we're not breathing and it builds up, we get a respiratory acidosis. Likewise, if we blow it off, we'll have respiratory alkalosis (because we're getting rid of an acid).
- Bicarb is basic, so if it builds up, we get a metabolic alkalosis. If bicarb is low, we have a metabolic acidosis.

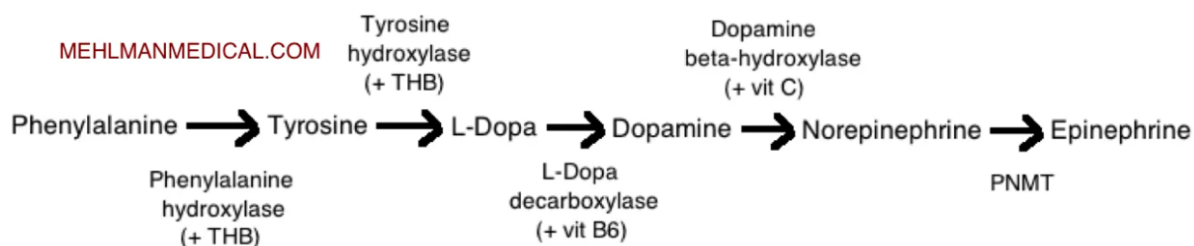
- **If pH is low:**

- **High CO_2 + normal HCO_3 = acute respiratory acidosis.**
- **High CO_2 + high HCO_3 = chronic respiratory acidosis (aka, respiratory acidosis with metabolic compensation).**
- **Low HCO_3 + low CO_2 = acute or chronic metabolic acidosis with respiratory compensation.**

- Primary metabolic acid/base derangements are compensated for instantly by changes in respiration (i.e., it's easy to retain or breathe off CO₂ merely by changing respiratory rate), whereas primary respiratory derangements are compensated for **slowly** because it takes time for the kidney to alter bicarb excretion.
 - Therefore, if CO₂ is up or down as the primary issue, a normal HCO₃ tells you the problem is acute (i.e., usually within the first day). If bicarb is not in the normal range, then it tells you the issue is chronic.
- **If pH is high:**
- **Low CO₂ + normal HCO₃** = acute respiratory alkalosis.
 - **Low CO₂ + low HCO₃** = chronic respiratory alkalosis (aka, respiratory alkalosis with metabolic compensation).
 - **High HCO₃ + high CO₂** = acute or chronic metabolic alkalosis with respiratory compensation.
- **Metabolic acidosis:**
- Low pH caused by low HCO₃; pCO₂ will instantly go down to compensate (CO₂ is acidic) by merely breathing more.
 - **MUDPILES** is mnemonic for high-anion-gap metabolic acidoses:
 - **M**ethanol, **U**remia (renal failure). **D**KA. **P**ropylene glycol / **P**aracetamol. **I**ron tablets + **I**soniazid. **L**actic acidosis. **E**thylene glycol (antifreeze). **S**alicylates (aspirin).
 - If the patient has metabolic acidosis and the anion gap is normal in the Q, you can eliminate the MUDPILES answers and choose an answer like renal tubular acidosis.
 - Aspirin is special in that it causes a mixed metabolic acidosis / respiratory alkalosis.
 - Renal tubular acidosis, Addison, and diarrhea are HY for normal anion gap.
 - Anion gap is calculated as $Na - (HCO_3 + Cl)$. Normal is 8-12 mEq/L. High is 13 or greater.
- **Metabolic alkalosis:**
- High pH caused by high HCO₃; pCO₂ will instantly go up to compensate (CO₂ is acidic) by merely breathing less. HY causes are:
 - Conn syndrome --> aldosterone-secreting tumor means more protons are secreted distally at the cortical collecting duct in the kidney.

- Renal artery stenosis + fibromuscular dysplasia --> RAAS is up so aldosterone is up.
 - Vomiting --> loss of protons in vomitus + volume depletion results in increased RAAS, further kicking out protons distally in the kidney.
 - Loops and thiazides --> cause "contraction alkalosis," which means volume depletion that causes RAAS to increase --> aldosterone goes up --> more protons secreted distally in the kidney --> metabolic alkalosis.
- **Respiratory acidosis:**
- Low pH caused by high CO₂; HCO₃ goes up *only if chronic* to compensate (bicarb is basic). It takes time for the kidneys to retain bicarb, so for the first 12-24 hours, bicarb will be normal.
 - Acute (**normal** bicarb) HY examples:
 - Hypoventilation due to opioids, benzos, barbiturates.
 - Can also be ventilatory insufficiency due to insufficient ventilator settings, or decompensation while on a ventilator.
 - Chronic (high bicarb) HY examples:
 - COPD --> these patients are chronic CO₂ retainers. Should be noted that *exacerbations of COPD* are acute on chronic respiratory acidosis, where CO₂ will be super high, with the bicarb already high.
 -
 - Obstructive sleep apnea / obesity hypoventilation syndrome --> chronic CO₂ retainers.
 - Ankylosing spondylitis + severe kyphoscoliosis --> lesser chest wall expansion.
- **Respiratory alkalosis:**
- High pH caused by low CO₂; HCO₃ goes down *only if chronic* to compensate (bicarb is basic). It takes time for the kidneys to excrete bicarb, so the first 12-24 hours, bicarb will be normal.
 - Acute (**normal** bicarb) HY examples:
 - Asthma attack --> CO₂ is low, not high. This is really HY for USMLE. CO₂ only normalizes or goes up if the patient gets tired, which is a danger scenario usually requiring intubation.

- Panic attack --> patient is hyperventilating.
- Most pulmonary emboli --> patient is hyperventilating. A low pH in the setting of PE can be seen in Qs, but this is due to lactic acidosis secondary to an obstructive shock process, not the PE directly.
- Altitude sickness (first day at high altitude) --> patient is hyperventilating.
- Chronic (**low** bicarb) HY examples:
 - Living at high altitude; pregnancy (progesterone upregulates respiratory centers).
- **Winter's formula:**
 - Winter formula is used to calculate predicted pCO₂ based on any change in bicarb, where:
 - Predicted pCO₂ = (1.5 x HCO₃) + 8 +/- 2.
 - For example, if a patient has DKA and bicarb is 14 mEq/L, we expect pCO₂ to be 27-31 mmHg (1.5 x 14 = 21; 21 + 8 = 29; 29 +/- 2 = 27-31).
 - If pCO₂ is lower than 27, then the patient has a concurrent respiratory alkalosis.
 - If pCO₂ is higher than 31, then the patient has a concurrent respiratory acidosis.
 - This type of Q can occasionally show up on USMLE, where if pCO₂ is in the expected range, the Q can write the answer as "compensated metabolic acidosis/alkalosis." If the change in CO₂ is less than what we expect, the Q can write the answer as "uncompensated metabolic acidosis/alkalosis."
- **Catecholamine synthesis:**



- Catecholamines are products derived from tyrosine that can act as both a hormone and a neurotransmitter. The adrenal medulla and CNS neurons are the two locations these are produced.
- **Phenylalanine --> tyrosine:**

- Phenylalanine hydroxylase (or THB) is deficient in **phenylketonuria (PKU)**. This is tested for via heelprick test at birth because not catching the disorder can cause mental retardation. Children with PKU present with partial albinism, mousy/musty body odor, and cognitive disability. The Q can say the child is fairer skinned than their sibling. Treatment is special diet without phenylalanine, since buildup of this amino acid causes symptoms. Tyrosine becomes an essential amino acid for the patient since they cannot synthesize it.
- **Tyrosine --> L-DOPA:**
- Tyrosine hydroxylase converts tyrosine --> L-DOPA. It is very rare for the USMLE to directly assess this enzyme. If it does, the answer will be obvious - i.e., "Patient has low dopamine synthesis secondary to an enzyme deficiency. What's the enzyme?" --> "deficiency of tyrosine hydroxylase," where L-DOPA decarboxylase isn't listed.
- Classic Parkinson disease is not due to deficiency of tyrosine hydroxylase, but instead due to loss of dopamine-secreting neurons in the substantia nigra pars compacta of the midbrain. But this enzyme deficiency can very rarely be a cause of Parkinsonism.
- Do not confuse tyrosine hydroxylase with **tyrosinase**. The latter is a HY cause of albinism (oculocutaneous albinism type I). You don't need to memorize this specific albinism type name, *but you must know that albinism is often due to enzyme deficiency in general*, where the patient cannot synthesize melanin. Tyrosinase converts tyrosine --> DOPA (not L-DOPA) --> dopaquinone in melanin synthesis. The USMLE can give you a vignette of albinism, and then the answer is just simply "enzyme deficiency," or "tyrosinase deficiency." The caveat, however, is that there is an NBME question floating around with a kid with red hair and partial albinism, where tyrosinase deficiency is wrong, but they show you a flow chart of chemical substrates, where it's obvious based on the info they give you that tyrosinase is wrong for that Q. There are numerous types of albinism with different enzyme deficiencies. USMLE probably made this Q so that you don't blanket-memorize tyrosinase deficiency as a 100% cause.
- **L-DOPA --> Dopamine:**
- L-Dopa decarboxylase is vitamin B6-dependent. This isn't dramatic if you remember that decarboxylase enzymes are often B6-dependent. Similar to tyrosine hydroxylase, deficiency of L-DOPA decarboxylase can result in dopamine deficiency and Parkinsonism.

- Patients with Parkinson disease aren't given dopamine directly, since dopamine has cardiovascular effects. So they are given L-DOPA/Carbidopa. For USMLE, they want you to know this combo "increases BBB penetration of L-DOPA." This is because enzymes that act on L-DOPA will act on carbidopa instead. The term "competitive inhibitor" isn't traditionally applied to carbidopa, but its presence increases the survival of L-DOPA so that it can cross the BBB.
- **Dopamine --> norepinephrine:**
- Requires dopamine beta-hydroxylase. For USMLE, you need to know this is **vitamin C-dependent**. There is NBME Q where they tell you patient has a buildup of dopamine and low NE, and the answer is simply "deficiency of dopamine beta-hydroxylase." Very easy question, but it exists.
- **Norepinephrine --> epinephrine:**
- The adrenal medulla is special in that it can convert NE --> E via an enzyme called phenylethanolamine-*N*-methyltransferase (PNMT). You actually do need to know the name of this enzyme for USMLE. This enzyme is upregulated by cortisol. Cortisol diffuses from the zona fasciculata of the adrenal cortex down to the medulla, where it upregulates PNMT synthesis.
- USMLE might tell you a patient's serum ratio of E:NE is 80:20 and ask why, and the answer is just simply "phenylethanolamine-*N*-methyltransferase," or "stimulation of epinephrine production by cortisol." The conversion of NE --> adds a methyl group, as implied by the enzyme name. You also need to know this.
- The difference between norepinephrine and epinephrine is that E can agonize beta-2 receptors, whereas NE cannot. This means if we block alpha-1 receptors with, e.g., phentolamine, and then give epinephrine, the beta-2 agonism results in peripheral vasodilation and a drop in blood pressure. This effect does not occur with norepinephrine. You need to know beta-2 agonism causes peripheral dilation and alpha-1 agonism causes peripheral vasoconstriction.
- NE and E are high in **pheochromocytoma**, where a patient will present with paroxysmal (episodic) hypertension, palpitations, and pounding headaches. The patient can sometimes have a BP of 120/80, and then when stressed (e.g., when sitting up on the operating table), have a BP that shoots up to 200/110. Pheochromocytoma can occur idiopathically, or be part of syndromes like NF1 or MEN2A/2B. Treatment is phenoxybenzamine, an irreversible alpha-1 blocker. Do not give a beta-blocker first for treatment, as this will kill the patient. If we block beta receptors first, then we cause

"unopposed alpha" binding, where the high amounts of NE and E floating around will bind to alpha-1, causing lots of arteriolar constriction, shooting BP higher and killing the patient.

- **Alkaptonuria:**

- Deficiency of homogentisic acid oxidase, leading to inability to breakdown both phenylalanine and tyrosine. This will cause blue/black discoloration of cartilage (ochronosis) and arthritis. The USMLE can tell you a patient has ochronosis and urine that turns black with time, or with the addition of citrate. Alkaptonuria is fairly obscure and isn't "HY" per se. But it's pass-level if it shows up because the presentation is so hyper-buzzy on the exam.
- 43M + gets sore knees when plays basketball + pinna of left ear slightly greyish + urine sample turns dark color upon standing; what enzyme is deficient in this patient? → answer = homogentisic acid (homogentisate) oxidase; required for tyrosine breakdown into fumarate. Sometimes Q will ask which amino acid is not broken down properly; answer = tyrosine.

- **Phakomatoses:**

- 20M + 6-year-Hx of seizure disorder + flesh-colored papules on bridge of nose and in the nasolabial folds; diagnosis? → answer = tuberous sclerosis (TSC); AD; chromosomes 9 and 16 (hamartin and tuberin proteins); subependymal or periventricular nodules seen on MRI; hypopigmented macules (ash leaf spots); hyperpigmented velvety lesions (shagreen patches); cardiac rhabdomyoma; renal angiomyolipoma; subungual fibromas (nailbed tumors); vignette will also give a young child who has writhing movements in his/her sleep to suggest seizures.

Phakomatoses (neurocutaneous disorders)	
Condition	HY points
Neurofibromatosis type I (NF1)	- <i>NF1</i> gene; AD; chromosome 17. - Neurofibromas, café au lait spots (hyperpigmented macules), axillary/groin freckling, Lisch nodules (iris hamartomas), pheochromocytoma, optic nerve glioma, oligodendroglioma, ependymoma.
Neurofibromatosis type II (NF2)	- <i>NF2</i> gene; AD; chromosome 22. - Bilateral acoustic schwannomas, meningioma.
Tuberous sclerosis (TSC)	- <i>TSC1/2</i> genes; autosomal dominant. - Periventricular nodules (tubers) on MRI of head, seizures, adenoma sebaceum (angiofibromas), subungual fibromas, renal angiomyolipoma, cardiac rhabdomyoma, hypopigmented macules ("ashleaf spots"), hyperpigmented velvety lesions ("shagreen patches").
Von Hippel-Lindau	- <i>VHL</i> gene; AD; chromosome 3. - Cerebellar/retinal hemangioblastomas, bilateral renal cell carcinoma, pancreatic cysts.

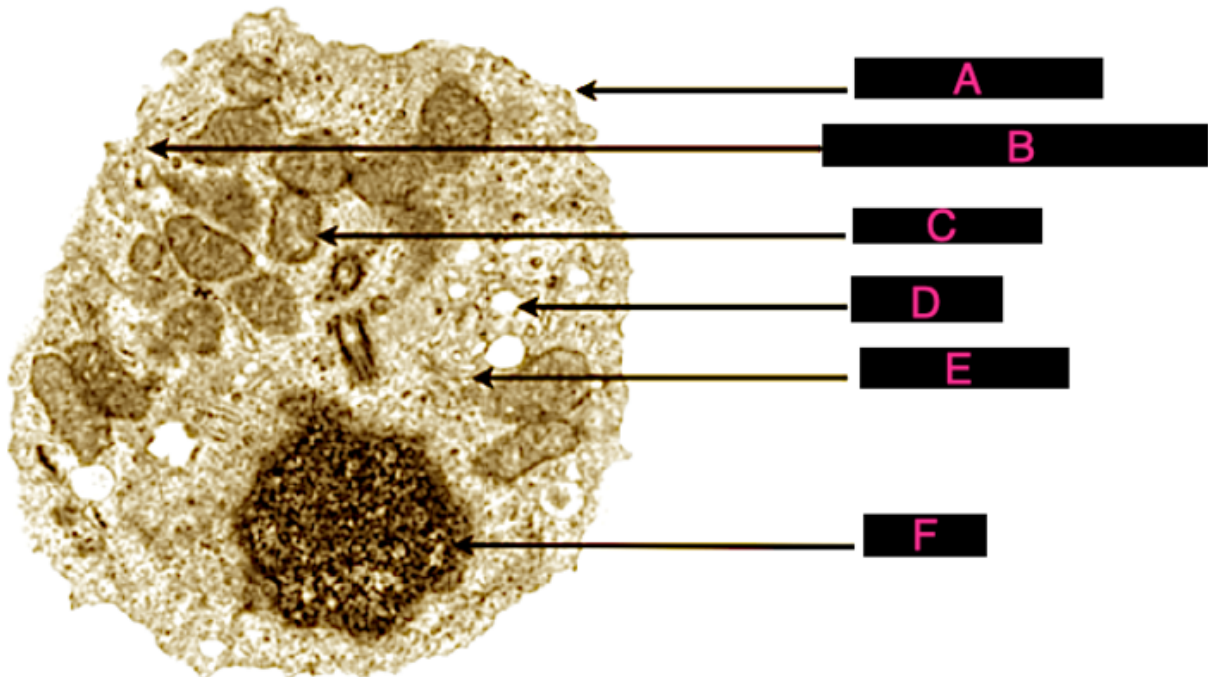
	- Mutation causes constitutive activation of hypoxia-inducible factor.
Sturge-Weber	- Somatic mosaicism condition; not inherited. - Port wine stain birth mark (nevus flammeus; may also present as violaceous papules in temporal distribution), leptomeningeal angioma (presenting as seizure); glaucoma

- Experiment is performed showing a nutrient causes increased gene products detectable by Western blotting + no change in gene products detected by Northern blotting or PCR; Q asks, what does the nutrient cause intracellularly? → answer = “translation of mRNA”; Western blot detects protein; Northern blot detects RNA; Southern blot detects DNA; PCR detects DNA. Transcription is the process of DNA → RNA; translation is the process of RNA → protein.

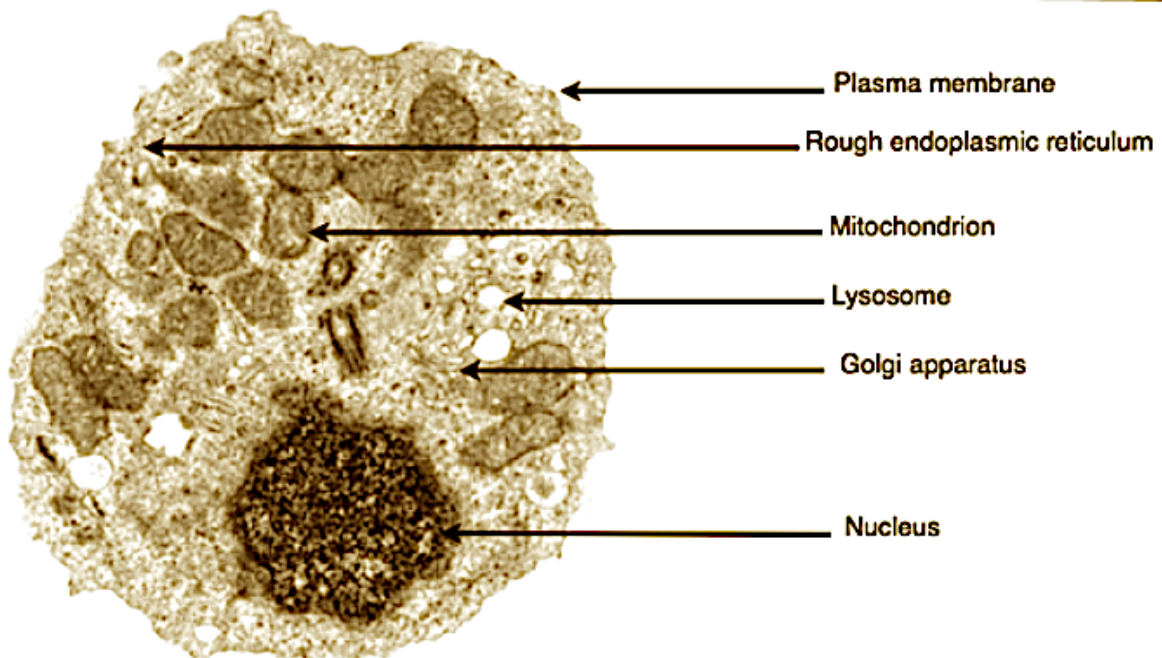
Annoying lab techniques	
Northern blot	- Detects RNA.
Southern blot	- Detects DNA.
Western blot	- Detects protein.
Southwestern blot	- Detects DNA-binding proteins.
ELISA	- Detects presence of antigen (direct ELISA) or antibody (indirect ELISA) in patient sample.
PCR	- Amplifies DNA fragments in order to make many copies.

- Experiment is performed visualizing changes to RNA via electron microscopy; Q asks which change occurs in the cytosol? → answer = “association with P-bodies”; DNA is transcribed to hnRNA (heteronuclear RNA); hnRNA then undergoes three main changes in the nucleus: 1) addition of poly-A tail to 3' end, 2) addition of 7-methylguanosine cap to 5' end, and 3) splicing out of introns; after these three changes are made in the nucleus, the hnRNA is now called mRNA (messenger RNA). The mRNA then leaves the nucleus for the cytoplasm, where it associates with P-bodies, which are “docking centers” for mRNA to be sequestered and translated at a later point in time.
- 28F + experiences spontaneous abortion; Q asks, what's the most likely mechanism for this? → answer = “unbalanced chromosomal rearrangement,” which is the most common cause of spontaneous abortion.
- Investigator looking at cellular effects of increased serum insulin; which of the following combinations of findings is most likely; answer = ↑ nuclear/cytoplasmic shuttling; ↑ serine phosphorylation; ↓ ubiquitination; insulin activates MAP tyrosine kinase and serine/threonine pathways, which will cause ↑ nuclear/cytoplasmic shuttling and ↑ serine phosphorylation; ubiquitination is ↓ because this is a catabolic process but insulin is anabolic.

- 20M + weakness + twitching + muscle biopsy shows ragged red fibers; Q asks which cellular organelle on the following diagram is fucked up.



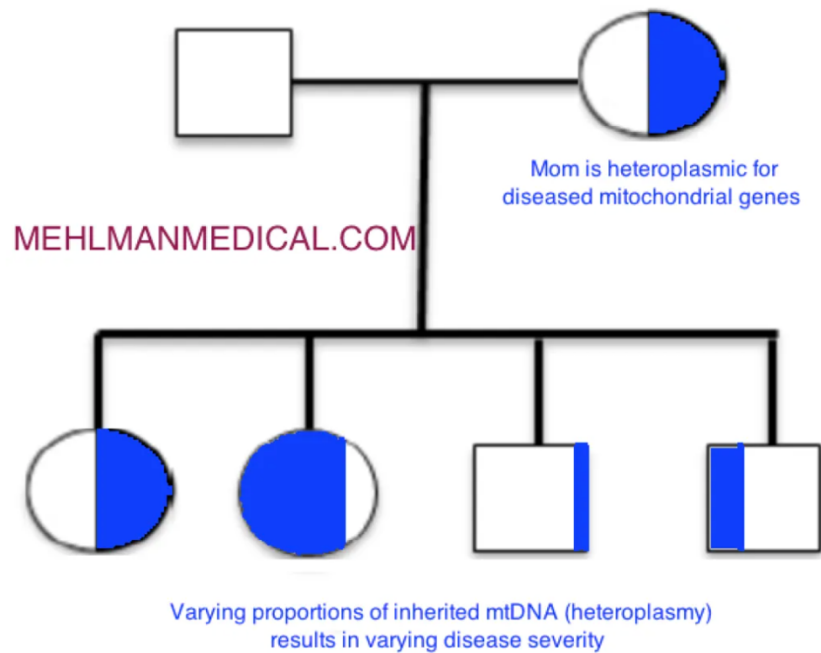
- Answer = mitochondrion; “ragged red fibers” is buzzy description seen in myoclonic epilepsy with ragged red fibers (MERRF syndrome).



- 4M + Hx of recurrent ear infections + hepatosplenomegaly + studies show deficient *N*-acetylglucosamine-1-phosphotransferase + analysis of skin fibroblasts shows they secrete large

amounts of acid hydrolase into the culture medium; Q asks, these hydrolase enzymes are unable to target to which organelle? → answer = lysosome; diagnosis is I-cell disease; deficient enzyme leads to inability to produce mannose-6-phosphate, which normally enables hydrolases at the Golgi to be targeted to the lysosomes; if the Q asks for the location of the deficient enzyme, choose Golgi; if the Q asks for the location the hydrolases cannot be targeted to, choose lysosomes. Presentation is normally restricted joint movement + coarse facial features, but can also present as recurrent ear infections and hepatosplenomegaly.

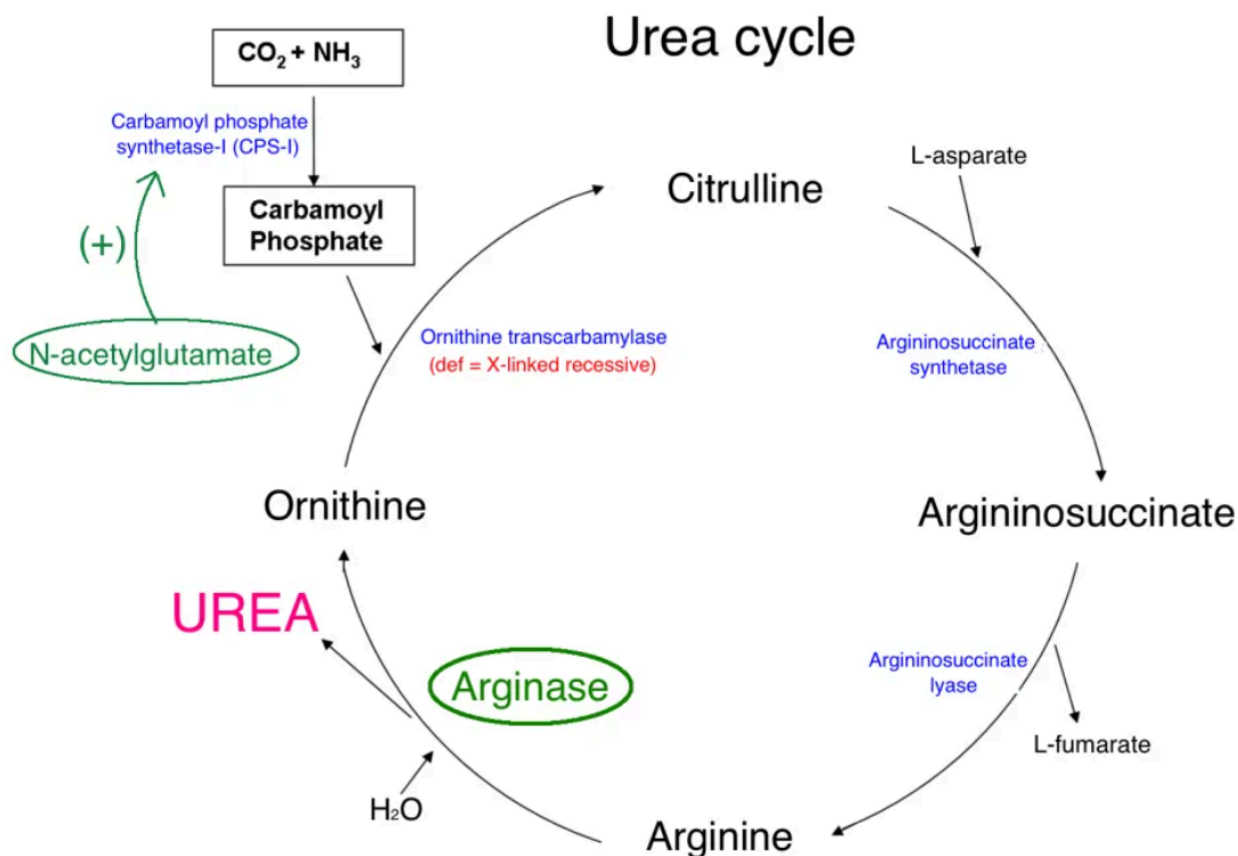
- 3M + developmental delay + ophthalmoplegia + hypotonia + hearing loss in mother + uncle had stroke-like episodes in his 20s; Q asks, which mechanism explains the findings in this family? → answer = heteroplasmy; diagnosis is a mitochondrial disorder (in this case, MELAS, which is Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like episodes); you don't need to know MELAS specifically, but just know ear and/or eye problems, lactic acidosis, and hypotonia suggest mitochondrial disorders in general; mitochondrial disorders are always maternally inherited (i.e., only females pass on to offspring); heteroplasmy is variation of mitochondrial disease severity based on the proportion of diseased mitochondrial DNA (not chromosomal DNA) inherited; each oocyte from the mom will have hundreds to thousands of copies of mitochondrial DNA with differing proportions of diseased vs healthy copies, leading to offspring with varying disease severity. Heteroplasmy is specifically applied to this process for mitochondrial diseases; in contrast, variable expressivity is applied to disorders with complete penetrance that are *not* mitochondrial (e.g., NF1).



- Neonate + marked hypopigmentation + blue irides; Q asks, this condition is caused by a defective enzyme located in which organelle? → answer = melanosome; diagnosis is albinism, which is usually caused by deficient/defective tyrosinase, a melanosomal enzyme.
- 27M + severe muscle cramping with exercise + venous blood sampling shows post-exercise lactic acid does not increase compared to pre-exercise levels; Q asks, which enzyme is most likely deficient? → answer = muscle glycogen phosphorylase; diagnosis is McArdle syndrome (glycogen storage disease type V).
- 16F + gives birth to neonate with epicanthal folds, slanted palpebral fissures, single palmar crease, and low IQ; karyotyping of the neonate shows abnormality in 100% of cells; what's the most likely mechanism for her child's condition? → answer = Robertsonian translocation; child has Down syndrome (trisomy 21) due to inheriting the long arm of chromosome 21 attached to chromosome 14 of one of the parents – i.e., $t(14q;21q)$. The implication is that it is unlikely to have meiotic nondisjunction in the female at such a young age (1/100 chance at age 40; 1/10 chance at age 50); if child is mosaic Down, then mechanism is “post-fertilization mitotic error” (i.e., error occurred during cell division after fertilization). Robertsonian accounts for 4% of Down, mosaicism 1%, nondisjunction 95%. Down syndrome → epicanthal folds, slanted palpebral fissures, flattened facies, single palmar crease, and low IQ, endocardial cushion defects (AVSD), pulmonary arterial malformations, hypothyroidism, Eustachian tube atresia (recurrent otitis media + hearing loss), Brushfield spots

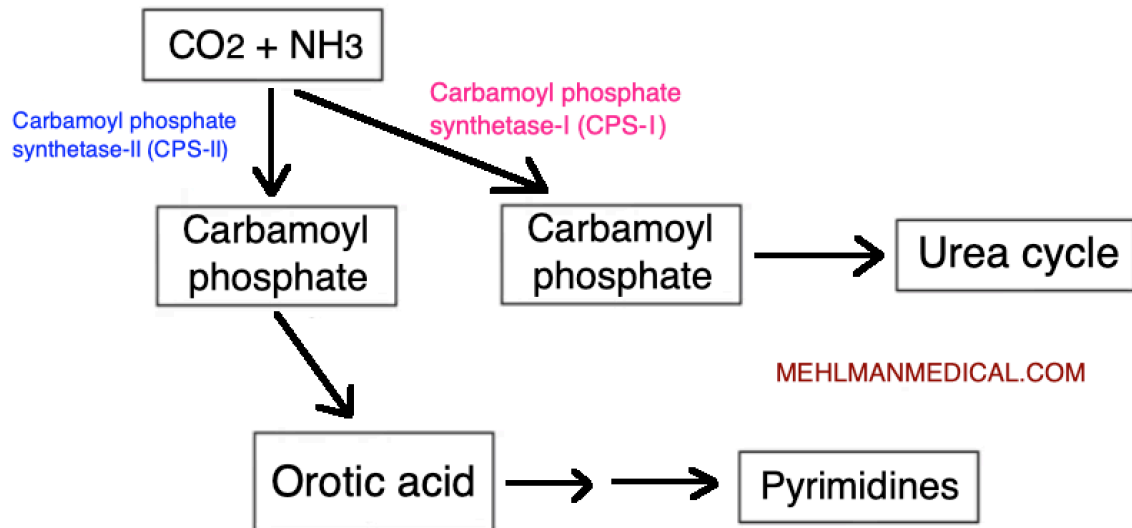
(white spots found on periphery of iris); tri-screen in first trimester of pregnancy at 8-10 weeks shows ↓ PAPP-A, ↑ β-hCG, ↑ nuchal translucency; ultrasound shows hypoplastic nasal bone (flattened facies); quad screen in second trimester of pregnancy at 16-20 weeks shows ↓ AFP, ↑ β-hCG, ↓ estriol, ↑ inhibin-A (two ways to remember: 1) put the variables in alphabetical order, then just know it's down, up, down, up; OR, 2) the variables that have an "H" in them are the ones that have ↑ arrows).

- Neonate born with microcephaly + prominent occiput + clenched hands + low-set ears; what chromosomal abnormality is most likely to be seen? → answer = trisomy of chromosome 18; diagnosis is Edward syndrome; presentation is microcephaly + prominent occiput + clenched hands + low-set ears; flattened facies **not** characteristic (that's Down due to hypoplastic nasal bone); second trimester quad screen shows ↓ for all four variables: AFP, β-hCG, estriol, inhibin-A.
- Neonate born with holoprosencephaly + cleft lip; Q asks about chromosomal abnormality; answer = trisomy of chromosome 13; Dx is Patau syndrome.
- 42F + has a tumor demonstrating G-alpha-s G-proteins that lack GTPase function; Q asks, which cellular change would be seen in this patient? → answer = increased cAMP; cleavage of GTP into GDP functions to shut off G-protein activity, so loss of GTPase function means increased activity; G-alpha-s G-proteins increase cAMP, so in this case the tumor leads to increased cAMP production.
- Neonate with rare disease has a mutation localized to an intron sequence; Q asks, which of the following mechanisms most likely accounts for disease in this patient? → answer = "RNA splice error"; introns are sequences that are not translated into protein; exons are sequences that can be translated into protein; a HY USMLE point regarding introns is that they play a role in post-transcriptional RNA splicing.
- Neonate + high serum BUN + normal creatinine; a defect in ammoniogenesis; Q asks, which substrate is the most likely source of the ammonia; answer = glutamine; ammonia is produced from the breakdown of various amino acids with amino group side-chains; the ammonia is processed via the urea cycle into urea, which is much less toxic.



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- 21M + $\uparrow$ serum orotic acid + hyperammonemia + $\downarrow$ blood urea nitrogen + normal RBC MCV; Dx? $\rightarrow$ answer = ornithine transcarbamylase deficiency; urea cycle disorder; X-linked recessive; orotic acid is normally produced from carbomyl phosphate (via CPS-II) in the synthesis of pyrimidines; in ornithine transcarbamylase deficiency, since the urea cycle isn't working, carbamoyl phosphate is shunted toward orotic acid; hyperammonemia results from less conversion of NH_3 to urea; MCV is normal because pyrimidine synthesis is not impaired; blood urea nitrogen is decreased because less urea is being produced.
- Neonate + poor feeding + lethargy + vomiting + increased serum orotic acid + hyperammonemia; which enzyme is deficient; answer = ornithine transcarbamylase.
- 21M + $\uparrow$ serum orotic acid + $\uparrow$ RBC MCV + serum B9 and B12 levels are normal; Dx? $\rightarrow$ answer = orotic aciduria; caused by deficiency of uridine monophosphate synthase, an enzyme required for pyrimidine synthesis; MCV is increased because pyrimidine synthesis is impaired; ammonia is normal (in contrast to ornithine transcarbamylase deficiency) because the urea cycle is intact.



- 8-month-old girl + hyperammonemia + decreased serum citrulline + increased serum glutamine + increased serum orotic acid; which enzyme is deficient in this patient? → answer = ornithine transcarbamylase deficiency; should be noted this is X-linked recessive but Step 1 NBME gives the condition in a girl; apart from the inheritance pattern discrepancy making no sense, the only way to arrive at the answer is because the combination of vignette findings points toward the deficiency; if urea cycle is impaired, glutamine cannot liberate ammonia into the urea cycle (so glutamine and ammonia build up); citrulline is decreased simply because it is one of the intermediates of the urea cycle; orotic acid is increased because the ammonia is shunted down pyrimidine synthesis instead (orotic acid is one of the intermediates); wrong answer is deficiency of carbamoyl phosphate synthetase because this enzyme is required for orotic acid synthesis. It's a poor Q because the disease is XR and the presentation is in a girl, but it's literally what's on the NBME.
- 5M + progressive weakness and loss of muscle function + exam shows hypertrophy of calf muscles + serum CK is elevated; the most likely cause of this presentation is a defective protein in which area of muscle? → answer = cytoskeleton; diagnosis is Duchenne (XR); dystrophin protein (*DMD* gene) is required to connect muscle cell cytoskeleton to extracellular proteins; pseudohypertrophy results from fibroadipose tissue deposition; Gower maneuver classic for Duchenne; Duchenne is caused by frameshift or nonsense mutations, resulting in truncated, non-functional protein; Becker is less severe and presents in teenage or young adult years and is usually caused by missense mutations or insertions/deletions of entire codons.

- Girl with asthma is administered IV methylprednisolone, resulting in marked clinical improvement; what's the mechanism via which this agent resulted in improvement? → answer = "binding to cytoplasmic receptor, translocation to nucleus, and transcriptional activation of target genes"; most steroid hormones bind to intracytoplasmic receptors, as is the case additionally for androgens, estrogens, progestins, glucocorticoids, and mineralocorticoids; thyroid hormone (T3/T4) bind to intranuclear receptors.
- Experiment is performed analyzed nuclear factor-kappa B (NF-κB); Q asks about role of IκB in relation to NF-κB; answer = "releases NF-κB after undergoing phosphorylation"; outrageously dumb and pedantic, but it's on the NBME so just memorize it.
- Experiment is performed where an agent inhibits histone deacetylase; this drug most likely affects which cellular process? → answer = transcription; histones are highly compacted structures of heterochromatic DNA; process of acetylation normally activates and makes DNA euchromatic (i.e., transcribable); methylation normally silences and makes DNA heterochromatic (i.e., not transcribable); histone deacetylase ↑ silencing (i.e., ↓ acetylation) and ↑ heterochromatin.
- 42M + receives all-*trans* retinoic acid for Tx of AML M3 (APL); Q asks, which enzyme will be attracted to the retinoic acid receptor/promyelocytic leukemia fusion protein to form a pre-transcriptional complex? → answer = histone acetylase; wrong answer is histone *N*-methyltransferase; learning point for USMLE is that acetylation is usually associated with gene activation; methylation is usually associated with gene silencing. Epigenetics is the study of how gene expression can be modified without changes made to nucleic acid sequences (i.e., instead through methylation, acetylation, etc.).
- 80M + alcohol dependence + many ecchymoses + few teeth + hemorrhagic macules around hair follicles; what is most likely responsible for his poor wound healing? → answer = inadequate hydroxylation of collagen polypeptides; Dx is scurvy (vitamin C deficiency); vitamin C is necessary for hydroxylation of proline and lysine residues in developing collagen; perifollicular hemorrhages are a HY finding for scurvy; bleeding around gums, poor wound healing, and easy bruising are common.
- 30M + palpitations + weight loss + exophthalmos + tremors; Q asks, patient's presentation is most likely due to hormonal action on which of the following receptors? → answer = "nuclear/retinoid X binding to DNA"; thyroid hormones bind intranuclear.

- A couple who are both healthy have two children with achondroplasia; what's the most likely mechanism for this process? → answer = germline mosaicism; achondroplasia is an AD disorder but most cases are due to *de novo* mutations within one of the spermatogonia or oogonia of the parents (i.e., a % of germline cells are affected), but the somatic (body) cells of the parents are completely normal (i.e., this is *not* somatic mosaicism); germline mosaicism is important in explaining how it's possible for an individual to have an AD disorder, where 100% of their somatic cells are affected, despite both parents being both phenotypically and genotypically normal. In contrast, somatic mosaicism is due to a post-fertilization mitotic error and is the mechanism via which a patient could have a % of somatic cells normal and a % diseased (e.g., a mosaic Turner or Down patient).
- 24F + lost in wilderness for three weeks; Q asks how adipocytes play an important role in maintaining homeostasis; answer = "glucagon activates hormone-sensitive lipase"; HSL will liberate fatty acids from the adipocytes into the blood to be used as energy. Insulin does the opposite by activating lipoprotein lipase; this brings fatty acids from the blood into the adipocytes.
- 19M + mental retardation + long face + prominent ears + large testes; analysis of patient's DNA shows 400 CGG repeats in the *FMR1* gene 5' untranslated region; which of the following would most likely result from this expansion of nucleotide repeats? → answer = "decreased transcription"; Dx is Fragile X, which is due to TNR expansion >200 repeats of CGG; expansion of untranslated region results in hypermethylation of *FMR1* gene and prevents binding of transcription factors and RNA polymerase.
- 6-month-old boy + jaundice + cataracts + hepatomegaly; urine shows positive reaction to copper reduction test and negative reaction to a test agent that contains glucose oxidase; what enzyme is deficient in this patient? → answer = galactose-1-phosphate uridylyltransferase (classic galactosemia); if the Q mentioned cataracts alone, answer would be galactokinase deficiency; the galactose disorders appear shortly after birth because lactose in breast milk (or cow's milk) is broken down into glucose and galactose; in contrast, fructose disorders appear later when the child is started on juice, fruit, honey, and table sugar, where sucrose is broken down into glucose and fructose.
- Neonate has blue sclerae, hyperextensible joints, fragile bones, and dysplasia of his enamel; which mechanism accounts for the pleiotropic effects of osteogenesis imperfecta in this patient? → answer = "expression of defective gene in multiple tissues"; pleiotropy once again is one gene resulting in different unrelated phenotypes.

- 2F + cherry red spot on macula + deficiency of hexosaminidase A; this patient is most likely to have a buildup of what? → answer = GM2 ganglioside; Dx is Tay-Sachs.
- Investigator looking at mitochondrial tRNA and finds mitochondria encode their own tRNA; Q asks why? → answer = “mitochondria use a non-standard genetic code”; mitochondria have their own genes separate from chromosomal DNA.
- 17F + asks about breastfeeding and immunity for her baby; physician explains IgA moves through epithelial cells and is secreted into duct lumina; Q asks which process is best described here; answer = transcytosis, which is movement of substrates from outside one side of the cell, *through* the interior of the cell, and then out the other side of the cell; this is in contrast to exocytosis, which is merely secreting substrates from inside the cell to the outside; endocytosis is a broad term referring to the internalization of substrates from outside the cell to within; phagocytosis is the engulfment of solid particles; pinocytosis is the engulfment of liquid.
- 50F + positive PPD test + negative chest x-ray + commenced on prophylactic therapy; 6 months later patient develops decreased sensation to pinprick in all extremities; Dx? → answer = vitamin B6 deficiency secondary to isoniazid.
- 3F + tonic-clonic seizure; Q asks, the Tx for this condition mediates its effect via which type of molecular target; answer = “ligand-gated ion channels”; benzos, e.g., diazepam, are used first-line for status epilepticus; they agonize GABAA, which causes chloride to enter the neuron, hyperpolarizing it. GABA is normally the ligand that binds to its receptor, activating the chloride channel.
- 2M + bites his fingers and lips + orange-red crystals in his diaper; which enzyme is deficient in this patient? → answer = hypoxanthine-guanine phosphoribosyltransferase (HGPRT); Dx is Lesch-Nyhan syndrome; X-linked recessive; congenital cause of uric acid overproduction (gout); self-mutilation common; HGPRT is necessary for recycling of purines; if enzyme missing, hypoxanthine and guanine are shunted to uric acid; red-orange sand-like uric acid crystals seen in diaper.
- 2F + potbelly + hypotonia + large tongue; Dx? → answer = cretinism (congenital hypothyroidism); can lead to mental retardation (thyroid hormone needed for myelin synthesis); impaired bone growth.
- 2F + potbelly + fed gruel; Dx? → answer = “protein-calorie malnutrition” (kwashiorkor) – i.e., sufficient calories but insufficient protein.
- 2F + severely underweight with wasting; Dx? → answer = “total calorie malnutrition” (marasmus).

- 3M + severe combined immunodeficiency + receives gene therapy delivered via lysogenic phage + treatment successful; six months later he develops leukemia; why? → answer = virus incorporated itself next to oncogene; since the treatment was successful, we know the delivered gene was successfully incorporated and transcribed, so for there to be leukemia means an oncogene may have been nearby and erroneously also transcribed.
- 10M + recurrent nosebleeds + bleeding time 6 minutes + platelet count 200,000 + Hb 13.5 g/dL + facial telangiectasias; Q asks, what's the inheritance pattern of this condition? → answer = autosomal dominant; Dx is hereditary hemorrhagic telangiectasia; normal bleeding time and platelet count signify this is not immune (idiopathic) thrombocytopenic purpura (ITP).
- 33M + non-smoker + worsening lung function + father died of alcoholic cirrhosis + younger brother is a smoker and has advanced lung disease; Q asks, the deficient enzyme in this patient normally targets what? → answer = elastase; Dx is α 1-antitrypsin deficiency; codominant inheritance with varying degrees of severity; homozygous ZZ alleles cause severe presentation; α 1-antitrypsin, despite the name, breaks down elastase in the lungs; deficiency leads to *pan*-acinar emphysema (smoking causes centri-acinar); elastase also present in the liver, so cirrhosis common; presumption is father of this patient had greater susceptibility to alcoholic cirrhosis because of heterozygosity.
- 42M + 7-month-Hx of shortness of breath on exertion + ejection fraction of 70% + Hx of epistaxis since adolescence + patient's tongue and finger are shown; Q asks, what is cause of patient's presentation?



- Answer = "pulmonary arteriovenous shunting"; Dx is hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu); autosomal dominant; can cause GI bleeding (iron deficiency anemia; fatigue); Q will either show you a red dot on the tongue or finger (telangiectasia); can lead to high-output cardiac failure (EF >70%) in middle age due to AV shunting (as with this patient); vignette can also be pediatric.

- 56M + HbA1c of 12% + decreased sensation to pinprick up to the ankles; Q asks, what's the mechanism for myelinopathy in this patient? → answer = increased activity of aldose reductase; in diabetes, glucose moves into myelin sheaths and is converted to sorbitol via aldose reductase; sorbitol causes osmotic damage due to cellular swelling.
- 12F + seizures + loss of vision + weakness + serum lactic acid increased + mother and maternal grandmother have hearing loss; where is the patient's mutation? → answer = mitochondrial tRNA^{Glu}; the point being: the kid has a mitochondrial disorder (classic findings of lactic acidosis, eye/ear problems, hypotonia/weakness), and mitochondrial tRNA^{Glu} is the only answer that matches in this Q.
- Investigator looking at mutations in *NF1* gene. A wildtype (healthy) sequence of 5'-ATGCCCT-3' → mutated sequence 5'-ATG**G**CCCT-3'; a restriction enzyme is designed to cleave the mutated sequence; what sequence will the restriction enzyme cleave at? → answer = 5'-GGCC-3'; most restriction enzymes cleave at palindromic sites (GGCC is a palindrome), versus, e.g., GCCC is still part of the mutated sequence but is not a palindrome.
- Graph Q where the more protein A is bound to a plate, the less protein B is able to bind to the plate; Q asks why this is the case; answer = proteins A and B express the same epitope; epitope = part of an antigen that antibody attaches itself to; don't confuse with antibody idiotype (Fab) and isotype (Fc), which I talk about in detail in HY Immuno.
- 34F + breast cancer + BRCA mutation; Q asks what kind of DNA repair is fucked up; answer = recombinational dsDNA break repair. In contrast, mismatch repair is answer for Lynch syndrome (HNPCC); nucleotide excision repair is xeroderma pigmentosum.
- 69F + fractures right hip + two months later right leg has less muscle mass compared to left leg; which cellular process might explain these findings? → answer = polyubiquitination → atrophy entails protein breakdown, where ubiquitination of proteins targets them for the proteasome.
- 25M + undergoing chemo for leukemia; a drug is given to help prevent renal failure; Q ask which enzyme this drug inhibits; answer = xanthine oxidase → allopurinol or febuxostat to prevent tumor lysis syndrome.
- 1M + coarse facial features + hypotonia + increased serum lysosomal enzymes; which molecule is deficient in this patient? → answer = mannose-6-phosphate (easy for I-cell disease).

- Q shows you absurdly long sequence of DNA base pairs; exons are bolded; introns are un-bolded; mutation is shown within an un-bolded region (i.e., an intron); Q asks what effect this would most likely have; answer = disruption of normal splicing; mutations within introns are sometimes splice-site mutations; for USMLE, just think: introns = do not become protein + mutations are often splice-site.
- Q asks about molecular effect of oxytocin; answer = increases phosphoinositide hydrolysis; oxytocin is G-alpha-q G-protein, which increases IP3 (via hydrolysis of PIP2) and DAG.
- Q asks which of the following analyses can help determine monoclonality of neoplastic cells; answer = X-chromosome-linked isoenzymes; isoenzymes are enzymes with different amino acid sequences that catalyze the same chemical reaction; monoclonality refers to a cancer originating from one cell; investigation of G6PD X-linked isoenzymes is a known method for determining clonality of cell populations. Weird and obscure, I get it. But it's on the NBME.
- Bacterial strain produces a mutated protein that functions better at higher temperature; what kind of mutation accounts for this change in function; answer = missense mutation; polymorphisms resulting in differing function are usually missense – i.e., one amino acid is substituted for another.
- 40F + high-anion-gap metabolic acidosis + blurry vision + hangs around paint thinner for fun; which molecule is responsible for the patient's change in vision? → answer = formic acid; methanol → formaldehyde → formic acid; the latter can cause blindness with methanol toxicity; methanol is part of MUDPILES for high-anion-gap metabolic acidosis (Methanol, Uremia, DKA, Phenformin, Iron/INH, Lactic acidosis, Ethylene glycol, Salicylates).
- Q mentions experiment involving abnormal elastin in lab animals; which amino acid is most likely to be abnormal in these lab animals; answer = lysine (need to know elastin contains lots of lysine). Student says, "um ok..." I agree, dumb and pedantic, but it's on NBME.
- 82F + chronic alcoholism + tea and toast diet + blood around hair follicles + bruising; what process demonstrates decreased function in this patient? → answer = hydroxylation of proline; vitamin C is an essential cofactor for hydroxylation enzymes; required for collagen synthesis; perifollicular hemorrhages, bleeding gums, and easy bruisability are common findings; "tea and toast" diet is non-specific and can be seen in many vitamin deficiencies; 2CK FM shelf says tea and toast for vitamin C; NBME for Step 1 doesn't mention diet but says chronic alcoholism in a scurvy Q.

- 12M + cystic fibrosis + ataxia + bilateral diminished reflexes + weakness of lower extremities; which vitamin is deficient (B12 not listed); answer = vitamin E; CF leads to fat-soluble vitamin malabsorption; vitamin E deficiency can present with neuropathy; other vitamin deficiencies causing neuropathy: B6 (answer for isoniazid use); B12 (pernicious anemia, chronic gastritis, gastrectomy, terminal ileectomy, Crohn disease, vegetarian/veganism, *Diphyllobothrium latum* infection); B1 (dry beriberi).
- 30F + serum triglycerides 3200 mg/dL + serum appears milky; which enzyme is deficient in this patient? → answer = lipoprotein lipase (hyperchylomicronemia); if vignette also mentions lipase elevation and abdo pain, this is simply due to pancreatitis (i.e., the disease itself isn't causing lipase elevation); hyperchylomicronemia also caused by deficiency of apolipoprotein-CII.
- 24M + high serum glucose + necrolytic migratory erythema; Q asks which process is most likely upregulated in this patient; answer = oxidation of fatty acids; glucagonoma will force patient into catabolic state (i.e., decreasing relative amount of insulin); in contrast, quick review of some HY wrong answers:
 - “Decreased carnitine acyltransferase activity” → enzyme required for fatty acids to enter mitochondrion from cytosol, so this enzyme is catabolic and would be increased in activity.
 - “Synthesis of glycogen” → clearly anabolic process and would not be increased in predominance of glucagon.
 - “Increased acetyl-CoA carboxylase activity” → enzyme is anabolic and converts acetyl-CoA to malonyl-CoA, which is a precursor to fatty acid synthesis; this would not be upregulate in catabolic state; in addition, malonyl-CoA is the allosteric inhibitor of carnitine acyltransferase, meaning that if malonyl-CoA is increased, oxidation of fatty acids would be decreased, which is clearly not the case in the catabolic state.
 - “Decreased ketogenesis” → we would expect increased, not decreased, ketogenesis in catabolic state. Acetyl-CoAs liberated from FA breakdown are the precursors for ketones in the catabolic state.
- Guy starving in wilderness for three weeks + has decreased muscle mass; which of the following best explains his current status; answer = negative nitrogen balance; this means intake of nitrogen compounds (i.e., protein) is less than utilization by the body, leading to loss of muscle mass; in bodybuilding, a positive nitrogen balance ensures muscle growth.

- 2M + fractures at different stages of healing + genetic mutation showing a glycine residue in the collagen I gene becomes an alanine; what's the mechanism for this patient's condition? → answer = weakening of secondary structure of collagen; glycine is a proton; alanine is a methyl group; this means if alanine replaces glycine, there's less hydrogen bonding between sidechains in the collagen molecule (hydrogen bonding is important for secondary structure in proteins); I often see students memorizing "Ok, osteogenesis imperfecta is decreased secondary structure." But this isn't the point. The Q is assessing your understanding of amino acids, hydrogen bonding, and secondary structure.
- Q asks for important effect of niacin on patient with increased serum TGAs; answer = antagonizes hepatic VLDL-cholesterol secretion (i.e., decreases TGAs); niacin notably increases HDL (not listed as an answer for this Q + the Q asks about a patient who specifically has high TGAs); PPAR-alpha agonism refers to fibrates (upregulate lipoprotein lipase).
- 4M + recurrent pancreatitis + high serum cholesterol and TGAs + receives an injection of heparin and this restores his serum lipoprotein lipase activity; which of the following is decreased in this patient; answer = ApoC-II; familial hyperchylomicronemia.
- Q asks about how to counsel healthy patient on good nutrition; what should be part of this patient's diet? → answer = linoleic and linolenic acids (the two essential unsaturated fatty acids); linoleic is omega-6; linolenic is omega-3; both are C18; wrong answer is stearic acid (C18 saturated FA that shows up frequently as distractor for biochem NBME Qs).
- Q asks about cellular effect of *Diphtheria* toxin; answer = disrupts translation; inhibits elongation factor 2; (same as *Pseudomonas*).
- Q asks about impaired trafficking of vesicles into the Golgi; which would be seen on electron microscopy of the cell? → answer = rough endoplasmic reticulum dilatation; normally proteins are trafficked from RER → Golgi, so if this process cannot occur, the RER should dilate.
- Q asks about what kind of protein alteration will result in failure of protein transport within a cell; answer = deletion of residue from N-terminus; signal-recognition particles that enable cellular trafficking of proteins recognize the N-termini of proteins; if the N-terminus is disrupted on a protein, the SRP cannot target the protein to its correct cellular destination. Pedantic? Right. Well it's on NBME.
- Q asks about where the receptor for signal-recognition particles is located; answer = RER.

- Q asks about which amino acid is necessary in a healthy child's diet in order to maintain proper nutrition; answer = methionine (essential amino acid); essential amino acid; memorize for USMLE.
- 27F + lost in wilderness + normoglycemia even after 3 weeks; which amino acid is likely to contribute to normal serum glucose in this patient? → answer = alanine; the four main gluconeogenic substrates in humans are lactate, glycerol, alanine, glutamine.
- Acute myocardial infarction; Q asks about what we'd expect in the ischemic cells in terms of intracellular sodium, intracellular potassium, and intracellular calcium; answer = ↑ intracellular Na, ↓ intracellular potassium, ↑ intracellular Ca. Normally the Na/K-ATPase pumps out 3Na + pumps in 2K for every 1ATP utilized. If Na builds up within the cell, this indirectly inactivates the Na/Ca antiporter (Na in; Ca out), since the gradient is no longer favored for Na; this results in more Ca staying within the cell. This is the same mechanism via which digoxin works (i.e., inhibits Na/K-ATPase, resulting in indirect inactivation of Na/Ca-ATPase).
- 2M + Hx of viral, fungal, bacterial, protozoal infections; investigations show a buildup of dATP in the patient's cells; this buildup inhibits which of the following enzymes? → answer = ribonucleotide reductase; child has AR form of SCID, resulting in adenosine deaminase (ADA) deficiency; ADA deficiency causes a buildup of dATP, which inhibits ribonucleotide reductase (on Step 1 NBME).
- Girl eats a meal; what's the mechanism of insulin secretion in her? → answer = fusion of vesicle with cell membrane; insulin secretion requires ATP-mediated closure of K channel → K builds up in the cell → depolarization → causes Ca influx → vesicles filled with insulin fuse with plasma membrane → insulin exocytosed.
- 5F + cystic fibrosis; Q asks, defect in which of the following is responsible for her condition → answer = protein structure; the chloride channel has defective protein structure, resulting in sequestration within the RER of the cytosol (i.e., channel is *not* transported to cell surface).
- Q asks which substrate is required to synthesize glucosamine from fructose-6-phosphate; answer = glutamine (yes, this is asked as a one-liner); F6P + glutamine → glucosamine + glutamate; glucosamine is a precursor for glycoproteins, glycolipids, and proteoglycans.
- Two different Qs asking about G-proteins; first asks effect of albuterol; other asks effect of ranitidine;
 - Albuterol → beta-2 agonist → agonizes G-alpha-s → increased cAMP.

- Ranitidine → antagonizes histamine-2 → antagonizes G-alpha-s → decreased cAMP.
- TATA box is mutated within a gene; what is the result of this? → answer = decreased binding of RNA polymerase; TATA box is region within the promotor region where RNA polymerase II binds to initiate transcription.
- 54F + temporal arteritis + IV methylprednisolone administered; how does this drug bind? → answer = cytosolic receptor with translocation to the nucleus; most steroid hormones bind cytosolic receptors, followed by translocation to the nucleus where gene transcription is upregulated.
- Investigator looking at organism that has incurred RNA polymerase mutation; why will this not adversely affect the species survival → answer = transcriptional errors are not transmitted to progeny.
- Investigator looking at plasma-membrane-associated proteins vs those that are secreted from the cell; which of the following is most likely to be observed to be different between these types of proteins → answer = transmembrane domain (present in plasma-membrane-associated proteins).
- Neonate + jaundice + hypoglycemia + reducing sugars seen in urine; assay for which of the following is most likely to establish the diagnosis → answer = activity of galactose-1-phosphate uridylyltransferase (classic galactosemia).
- G-protein signaling is disrupted in a patient; the effect of which of the following is most likely to be attenuated? → answer = epinephrine (since agonizes adrenergic receptors, which are G-protein-coupled).
- 3% saline is administered to a patient; what will happen to both osmolality and volume within the ECF and ICF? → answer = ↑ ECF volume; ↑ ECF osmolality; ↓ ICF volume; ↑ ICF osmolality; 3% is hypertonic (isotonic = 0.9%); ↑ osmolality of ECF will cause fluid shift out of ICF compartment, thereby increasing osmolality of ICF.
 - Administration of hypotonic saline: ↑ ECF volume; ↓ ECF osmolality; ↑ ICF volume; ↓ ICF osmolality.
 - Administration of isotonic saline: ↑ ECF volume; no change ECF osmolality; no change ICF volume; no change ICF osmolality.

- Cortisol increases synthesis of which of the following enzymes? → answer = phenylethanolamine N-methyltransferase (converts NE → E); norepinephrine and cortisol share venous drainage from the medulla, where conversion to epinephrine occurs.
- Neonate has progressive lethargy + hypotonia + metabolic acidosis + hyperammonemia; what type of condition is most likely responsible? → answer = organic acid metabolism disorder; common type is methylmalonic acidemia; organic acids, such as methylmalonic acid, can disrupt urea cycle (↓ conversion of ammonia to urea); the acids themselves ↓ pH.
- How to overcome methotrexate toxicity? → leucovorin rescue (folinic acid; *not* folic acid).
- How to overcome 5-FU toxicity → oral thymidine or uridine.
- What two substrates are required for heme synthesis? → answer = glycine and succinyl-CoA (combine via B6 and delta-ALA synthase → delta-ALA as first step in heme synthesis).
- What best explains altered mental status in diabetic ketoacidosis? → answer = dehydration both intra- and extracellularly (polyuria causing fluid depletion).
- Investigator looking at a protein that serves as a precursor for various hormones; which of the following molecular processes is required in order to generate the hormones from the precursor protein? → answer = post-translational modification.
- Patient with terminal ileectomy; which of the following molecular processes is most likely to be disrupted? → answer = conversion of N⁵-methyltetrahydrofolate to tetrahydrofolate; vitamin B12 is required for conversion of uracil → thymine, as a methyl group is transferred to uracil in order to make thymine.
 - 1) Homocysteine + 5-methyl-THF, via B12 → methionine + THF
 - 2) Methionine + ATP → SAM
 - 3) SAM donates -CH₃ elsewhere → homocysteine
- Patient has peripheral neuropathy + normal MCV + decreased hemoglobin + increased LDH; Q asks for vitamin deficiency (B12 and E both listed); answer = vitamin E deficiency. Increased serum LDH can be seen with hemolysis / RBC turnover; vitamin E deficiency can lead to RBC fragility and hemolysis. B12 deficiency causes increase in MCV.

- Neonate + jaundice + increased LFTs + increased serum very-long-chain fatty acids; which organelle is the site of the defect? → answer = peroxisome (required for very-long-chain and branched-chain FA breakdown).
- Which phase of the cell cycle are mitotic cyclins synthesized? → answer = G2 (preparation for mitosis). Cell cycle phases are G1, S, G2, M. Interphase comprises G1, S, G2; mitotic phase is M.
 - G1 → cellular contents (excluding chromosomes) are duplicated.
 - S → DNA synthesis (46 chromosomes duplicated by the cell).
 - G2 → more cell growth + mitotic cyclin synthesis.
 - M → mitosis + cytokinesis; nuclear envelope proteins phosphorylated during cytokinesis.
- Neonate + lactic acidosis + ketonemia + hypoglycemia; fructose + glycerol fail to increase glucose levels; galactose does increase glucose levels; what kind of defect does this patient have? → answer = defect in gluconeogenesis; galactose $\leftrightarrow$ galactose-1-phosphate $\leftrightarrow$ galactose-6-phosphate $\leftrightarrow$ glucose-6-phosphate; the latter, via glucose-6-phosphatase → glucose.
 - Glycerol → dihydroxyacetone phosphate (DHAP) $\leftrightarrow$ F-1,6-BP $\rightarrow\rightarrow$ glucose.
 - Fructose → F-1-phosphate → dihydroxyacetone phosphate $\leftrightarrow$ F-1,6-BP $\rightarrow\rightarrow$ glucose.
 - You can see the entrance of galactose into glycolytic intermediates (i.e., G6P) is closer in proximity to glucose, so there must be a defect between DHAP and G6P in the aforementioned patient.



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